PHYSICS

1) A screw gauge has a pitch of 0.5 mm and number of divisions on its circular scale is 100. While measuring thickness of a sheet following data obtained:
(a) Linear scale reading = 12 divisions
(b) Zero error in instrument = +0.010 mm.
(c) Corrected reading obtained after eliminating zero error = 6.255 mm
Then, circular scale division which coincides with reference line is :

(1) 45th div
(2) 50th div
(3) 53rd div
(4) 57th div

2) A calorimeter contains 250g of water at 25°C. A metal block of 200g and specific heat 0.1 cal $g^{-1}°C^{-1}$ at 100°C is placed inside it. If the final temperature becomes 30°C, the water equivalent of calorimeter is.

(1) 20g
(2) 25g
(3) 30g
(4) 40g

3) In Searle's apparatus to measure the Young's modulus of material of a wire, the diameter of experimental wire when measured by a screw gauge of least count 0.001 cm is 0.100 cm and length of wire L found to be 157.0 cm when measured with a meter scale of least count 0.1 cm. A weight of 50N causes an extension of x = 0.050 cm when measured by a screw gauge of least count 0.001 cm. The maximum possible error in value of Young's modulus is :

(1) 1.5×10^9 N/m^2
(2) 3×10^9 N/m^2
(3) 4.5×10^9 N/m^2
(4) 8.1×10^9 N/m^2

4) To find the resistance of a galvanometer by half deflection method, a student observed following data as given in table below:

S.No.	Resistance R (Ω)	Deflection in galvanometer (θ)	Shunt resistance S(Ω)	Half deflection (θ/2)	Galvanometer resistance (G)
1	3000	30	96Ω	15	G_1
2	4900	30	98Ω	15	G_2

From above galvanometer resistance G is near to :-

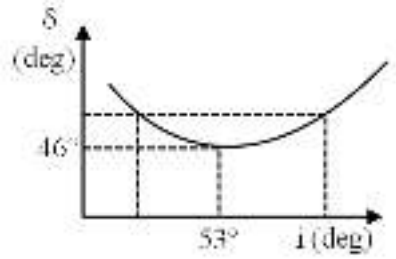
(1) 50Ω

(2) 80Ω

(3) 100Ω

(4) 120Ω

5) A graph between angle of incidence i and angle of deviation δ for a prism placed in air is as shown



in figure. The speed of light in prism is estimated near to :

(1) 1.5×10^8 ms^{-1}

(2) 1.75×10^8 ms^{-1}

(3) 1.9×10^8 ms^{-1}

(4) 2.5×10^8 ms^{-1}

6) To find out the refractive index of a glass slab using travelling microscope, which is having 20 divisions on 1 cm of its main scale and its 50 vernier scale divisions = 49 main scale divisions, following observations are made :
(A) For mark on paper seen without slab :
MSR = 8.25 cm ; VC = 15
(B) For mark on paper seen with slab :
MSR = 6.57 cm ; VC = 35
(C) For top surface of slab with powder :
MSR = 3.85 cm; VC = 5
(MSR = Main scale reading, VC = Vernier coinciding division)
Refractive index of glass slab is :

(1) 1.47

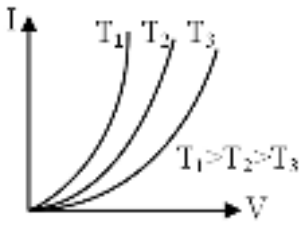
(2) 1.50

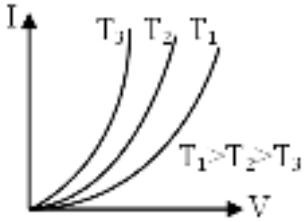
(3) 1.55

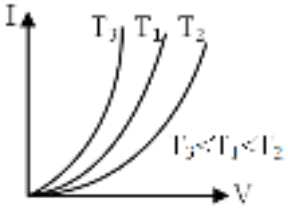
(4) 1.60

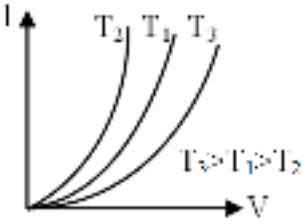
7) Which graph correctly represents the variation of P–N diode characteristics with temperature for forward bias.
(I → Current ; V → Applied voltage)

(1)

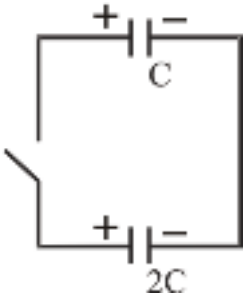


(2) 

(3) 

(4) 

8) Two capacitor are shown in the diagram. Each is charged to potential V_0. If capacitor C is filled with dielectric constant 2 and 2C is filled with dielectric constant 3 then key is closed. Final voltage on the capacitor C will be :-



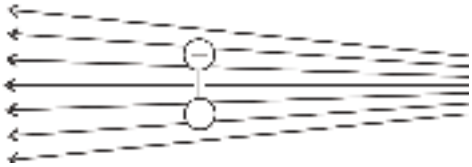
(1) $\frac{3}{8}V^0$

(2) $\frac{3}{5}V^0$

(3) $\frac{2}{5}V^0$

(4) V_0

9) When a dipole is placed in the non-uniform electric field shown below, the dipole is subjected to



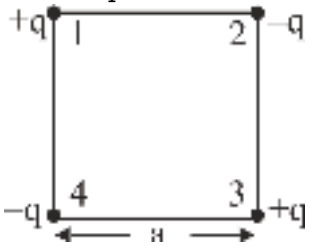
(1) a clockwise torque and a net linear acceleration

(2) a counterclockwise torque and a net linear acceleration

(3) a clockwise torque and no net linear acceleration

(4) a counterclockwise torque and no net linear acceleration

10) The work done required to put the four charges together at the corners of a square of side a, as shown in the figure is :



(1) $\frac{1}{4\pi\varepsilon_0}\frac{q^2}{a}$

(2) $-\frac{2.6}{4\pi\varepsilon_0}\frac{q^2}{a}$

(3) $+\frac{2.6}{4\pi\varepsilon_0}\frac{q^2}{a}$

(4) None of these

11) A proton is projected with a speed of 2×10^6 m/sec at angle 60° to x-axis. If a uniform magnetic field of 0.104 Tesla is applied along y-axis, the path of proton is

(1) A circle of radius 0.2m and time period $\pi \times 10^{-7}$ sec.

(2) A circle of radius 0.1m and time period $2\pi \times 10^{-7}$ sec

(3) A helix of radius 0.1m and time period $2\pi \times 10^{-7}$ sec

(4) A helix of radius 0.2m and time period $4\pi \times 10^{-7}$ sec

12) A specimen of iron of permeability $8 \times 10^{-3} \frac{Wb}{A-m}$ is placed in a magnetic field of intensity 160 A/m. Then magnetic field in this iron is :

(1) 20×10^3 Wb/m^2

(2) 1.28 Wb/m^2

(3) 5×10^{-5} Wb/m^2

(4) 0.8Wb/m^2

13) A conductor wire having 10^{29} free electrons/m^3 carries a current of 20 A. If the cross-section of the wire is 1mm^2, then the drift velocity of electrons will be :- ($e = 1.6 \times 10^{-19}$ C)

(1) 1.25×10^{-5} ms^{-1}

(2) 6.25×10^{-3} ms^{-1}

(3) 1.25×10^{-4} ms^{-1}

(4) 1.25×10^{-3} ms^{-1}

14) A current I = 10sin(100 πt) A is passed in first coil, which induces a maximum emf of 5 πV in

second coil. The mutual inductance between the coils is :-

(1) 5 mH
(2) 10 mH
(3) 15 mH
(4) 20 mH

15) The Q factor of a series LCR circuit with L = 2H, C = 32μF and R = 10 Ω is :-

(1) 15
(2) 20
(3) 25
(4) 30

16) **Statement-I :** Infrared waves are sometimes referred as heat waves.
Statement-II : Infrared waves heat up the earth's surface.

(1) Both statement-I and statement-II are true.
(2) Both statement-I and statement-II are false.
(3) Statement-I is true but statement-II is false.
(4) Statement-I is false but statement-II is true.

17) A microscope has an objective of focal length 1.5 cm and an eye-piece of focal length 2.5 cm. If the distance between objective and eye-piece is 25 cm, the approximate value of magnification produced for relaxed eye is ?

(1) 75
(2) 110
(3) 140
(4) 25

18) In YDSE, an interference pattern is obtained for λ = 6000 Å, coming from two coherent sources S_1 and S_2. At certain point P on the screen third dark fringe is formed, then the path difference $S_2P–S_1P$ in micron is :-

(1) 0.75
(2) 1.5
(3) 3.0
(4) 4.5

19) Four polaroids are placed such that the optic axis of each is inclined at an angle of 30° with the optic axis of the preceding one. If unpolarised light of intensity I_0 falls on the first polaroid, the light transmitted from the fourth is :-

(1) I_0
(2) $I_0/16$

(3) 0.21 I_0

(4) $\frac{I_0}{2}$

20) In the Bohr model of H-atom, if v is speed of electron and r is its orbital radius, then in case of a hydrogen atom is :-

(1) $r = \frac{e^2}{8\pi\in_0 mr^2}$

(2) $r = \frac{e^2}{8\pi\in_0 r}$

(3) $r = \frac{e^2}{4\pi\in_0 mv^2}$

(4) $r = \frac{e^2}{4\pi\in_0 v}$

21) Find the numerical value of DeBroglie wavelength of an electron in the 1st orbit of hydrogen atom assuming Bohr's atomic model. You can use standard values of the constants.

(1) 0.53 π A°

(2) 0.27 π A°

(3) 0.81 π A°

(4) 1.058 π A°

22) If the work function for a certain metal is 3.2×10^{-19} joule and it is illuminated with light of frequency 8×10^{14} Hz, the maximum kinetic energy of the photoelectrons would be in joule :-

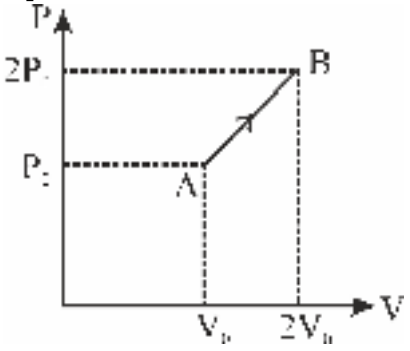
(1) 2.1×10^{-19}

(2) 8.5×10^{-19}

(3) 5.3×10^{-19}

(4) 3.2×10^{-19}

23) The P-V diagram of 2g of helium gas for a certain process A → B is shown in the figure. What is the heat given to the gas during the process A → B?



(1) $4P_0V_0$

(2) $6P_0V_0$

(3) $4.5P_0V_0$

(4) $2P_0V_0$

24) A gaseous mixture enclosed in a vessel consists of one mole of a gas A with (γ=5/3) and another B with (γ = 7/5) at a temperature T. The gases A and B do not react with each other and assumed to be ideal. Find the number of moles of the gas B, if γ for the gaseous mixture is (19/13).

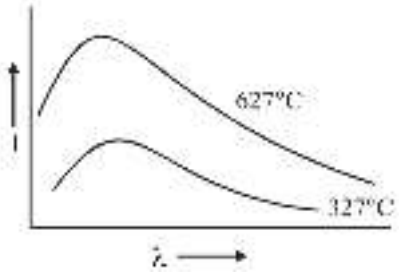
(1) 2

(2) 3

(3) 4

(4) 6

25) The spectra of a black body at temperatures 327°C and 627°C are shown in the fig. If A_1 and A_2 be the areas under the two curves respectively, the value of $\frac{A_1}{A_2}$ is :



(1) $\frac{81}{16}$

(2) $\frac{9}{4}$

(3) $\frac{27}{8}$

(4) $\frac{16}{81}$

26) Two rods of equal length and area of cross-section are kept parallel and lagged between temperature 20°C and 80°C. The ratio of the effective thermal conductivity to that of the first rod is : [K_1/K_2 = 3 : 4] :-

(1) 7 : 4

(2) 7 : 6

(3) 4 : 7

(4) 7 : 8

27) The volume coefficient of expansion of a liquid is 7 times the volume coefficient of expansion of the vessel. The ratio of absolute and apparent expansion coefficient of the liquid is :-

(1) $\frac{1}{7}$

(2) $\frac{7}{6}$

(3) $\frac{6}{7}$

(4) $\frac{1}{6}$

28) The potential energy of a particle executing S.H.M. along the x-axis is given by $U = U_0(1 - \cos \alpha x)$. What is the time period of oscillation? (for small αx)

(1) $2\pi\sqrt{\frac{m\alpha}{U_0}}$

(2) $2\pi\sqrt{\frac{U_0}{m\alpha}}$

(3) $\frac{2\pi}{\alpha}\sqrt{\frac{m}{U_0}}$

(4) $2\pi\sqrt{\frac{m}{\alpha U_0}}$

29) If the energy (E), velocity (v) and force (F) be taken as the fundamental quantity, then the dimensions of mass will be :

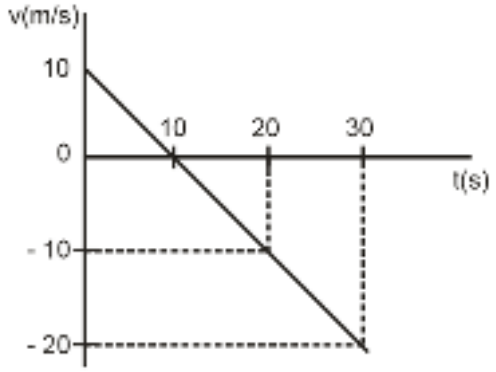
(1) Fv^{-2}

(2) Fv^{-1}

(3) Ev^{-2}

(4) Ev^{2}

30) The velocity-time plot for a particle moving on a straight line is shown in the figure.



(1) Acceleration of particle for 0 to10 sec is positive and for 10 to 30 sec is negative.

(2) The particle has never turned around.

(3) The particle has zero displacement in given time interval.

(4) The average speed in the interval 0 to 10 sec is the same as the average speed in interval 10 sec to 20 sec.

31) Figure represents a painter in a crate which hangs alongside a building. When this painter of mass 100 kg pulls the rope, the force exerted by him on the floor of the crate is 450 N. If the mass of



the crate is 25 kg, the acceleration of the painter will be (Take g = 10 m/s^2):-

(1) 1 m/s^2

(2) 2 m/s^2

(3) 3 m/s^2

(4) 4 m/s^2

32) A body of mass 40 kg resting on horizontal surface is subjected to a force P which is just enough to start the motion of body If μ_s = 0.5, μ_k = 0.4, g = 10 m/sec^2 and the force P is continuously applied on the body then the acceleration of the body is :-

(1) 0

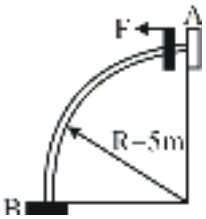
(2) 1 m/sec^2

(3) 2 m/sec^2

(4) 2.4 m/sec^2

33)

A bead of mass $\frac{1}{2}$ kg starts from rest from A to move in a vertical plane along a smooth fixed quarter ring of radius 5 m, under the action of a constant horizontal force F = 5 N as shown. The speed of bead as it reaches the point B is [Take g = 10 m/s^2] :-



(1) 14.14 m/s

(2) 7.07 m/s

(3) 5 m/s

(4) 10 m/s

34) A block of mass 2 kg is placed on a rough table at a distance 1 m from the center of table. The table starts rotating with constant angular acceleration of 4 rad/s^2 about an axis passing through its center. Find the centripetal acceleration of the block at t = 1 sec :

(1) 8 m/s^2

(2) 10 m/s^2

(3) 12 m/s^2

(4) 16 m/s^2

35) A ball falling freely from a height of 5 m, hits a horizontal surface. If $e = \frac{3}{4}$, then the time interval between first and second collision :-

(1) 1.0 s

(2) 1.5 s

(3) 2.0 s

(4) 3.0 s

36) A rope of negligible mass is wound round a hollow cylinder of mass 3 kg and radius 40 cm. If the rope is pulled with a force of 30 N then angular acceleration of cylinder is :-

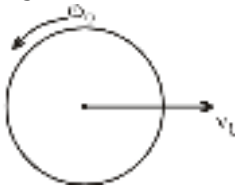
(1) 25 rad/sec^2

(2) 50 rad/sec^2

(3) 12.5 rad/sec^2

(4) 100 rad/sec^2

37) A uniform circular disc placed on a rough horizontal surface has initially a velocity v_0 and an angular velocity ω_0 as shown in the figure. The disc comes to rest after moving some distance in the direction of motion. Then $\frac{v_0}{r\omega_0}$ is :



(1) $\frac{1}{2}$

(2) 1

(3) $\frac{3}{2}$

(4) 2

38) A cylindrical tank has a hole of $1cm^2$ in its bottom. If the water is allowed to flow into the tank from a tube above it at the rate of 70 cm^3/sec then the maximum height upto which water can rise in the tank is :-

(1) 2.5 cm

(2) 5 cm

(3) 10 cm

(4) 0.25 cm

39) Two metal spheres are falling through a liquid of density 2×10^3 kg/m^3 with the same uniform speed. The material density of sphere 1 and sphere 2 are 8×10^3 kg/m^3 and 11×10^3 kg/m^3 respectively. The ratio of their radii is :-

(1) $\frac{11}{8}$

(2) $\sqrt{\frac{11}{8}}$

(3) $\frac{3}{2}$

(4) $\sqrt{\frac{3}{2}}$

40) A tunnel is dug along the diameter of the earth (radius R and mass M). There is a particle of mass 'm' at the centre of the tunnel. The minimum velocity given to the particle so that it just reaches to the surface of the earth is :-

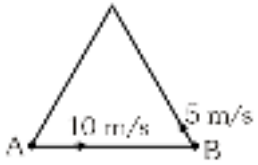
(1) $\sqrt{\frac{GM}{R}}$

(2) $\sqrt{\frac{GM}{2R}}$

(3) $\sqrt{\frac{2GM}{R}}$

(4) It will reach with the help of a negligible velocity

41) Two boys A & B are standing at the corners of equilateral triangle of side 5 m. They start moving along the sides with constant speed as shown. The time at which they meet for 2nd time is :-



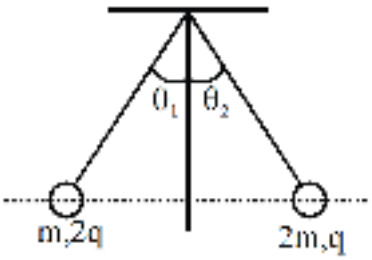
(1) 1 s

(2) 3 s

(3) 5 s

(4) 4 s

42) Two particles which have masses and charges m and 2q and 2m and q respectively are hung by massless threads from a point as shown in equilibrium particles are on same horizontal line. What can be the relation in θ_1 & θ_2. (θ_1 & θ_2 are not upto scale in figure) :



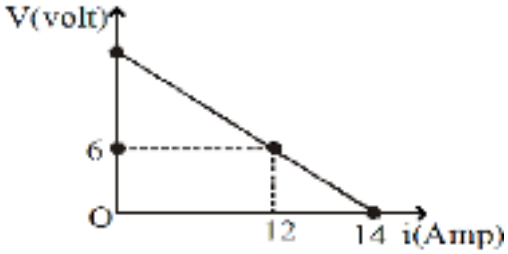
(1) $\theta_1 > \theta_2$

(2) $\theta_1 < \theta_2$

(3) $\theta_1 = \theta_2$

(4) More information is required.

43) 10 Cells, each of emf 'E' and internal resistance 'r', are connected in series to a variable external resistance. Figure shows the variation of terminal potential difference of their combination with the current drawn from the combination. Emf of each cell is :



(1) 1.6 V

(2) 3.6 V

(3) 1.4 V

(4) 4.2 V

44)

Consider the following nuclear reactions and select the correct statements from the options that follow.

Reaction I : $n \rightarrow p + e^- + \bar{\nu}$

Reaction II : $p \rightarrow n + e^+ + \nu$

(A) Free neutron has higher mass than proton, therefore reaction I is possible

(B) Free proton has less mass than neutron, therefore reaction II is not possible for the free proton

(C) Inside a nucleus, both decays (reaction I and II) are possible

(D) Inside a nucleus, reaction I is not possible but reaction II is possible.

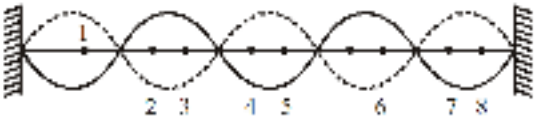
(1) A, B, C are correct

(2) A, C, D are correct

(3) A, B, D are correct

(4) B, C, D are correct

45) A stretched string is vibrating in its 5th harmonic as shown. Consider a particle 1 (figure). At an instant this particle is at mean positions and is moving towards its negative extreme. Which of the following set of particles, are in same phase with particle 1.



(1) 2, 4, 7

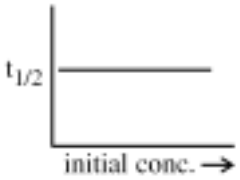
(2) 3, 5, 6

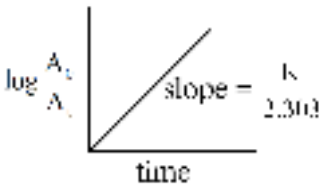
(3) 5, 7, 8

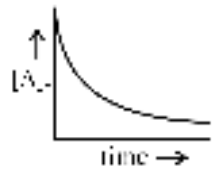
(4) 2, 4, 6

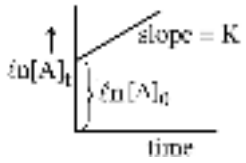
CHEMISTRY

1) Which of the following graph is incorrect for first order reaction ?

(1) 

(2) 

(3) 

(4) 

2) For the reaction 2A + 3B → Product, A is in excess and on changing the concentration of B from 0.1 M to 0.4 M rate becomes doubled. Thus rate law is :-

(1) $\frac{dx}{dt} = k[A]^2 [B]^2$

(2) $\frac{dx}{dt} = k[B]^{1/2}$

(3) $\frac{dx}{dt} = k[A] [B]$

(4) $\frac{dx}{dt} = k[A]^0 [B]^2$

3) The difference between n^{th} and $(n + 1)^{th}$ Bohr's radius of H atom is equal to 5 times of radius of 1st orbit. The value of n is :-

(1) 1

(2) 2

(3) 3

(4) 4

4) Equivalent conductance of $BaCl_2$, H_2SO_4 and HCl are x_1, x_2 and x_3 S cm^2 eq^{-1} at infinite dilution. If specific conductance of saturated $BaSO_4$ solution is Y S cm^{-1}, then K_{sp} of $BaSO_4$ is :

(1) $\frac{10^3 Y}{2(x_1 + x_2 - 2x_3)}$

(2) $\frac{10^6 Y^2}{(x_1 + x_2 - 2x_3)^2}$

(3) $\frac{10^6 Y^2}{4(x_1 + x_2 - x_3)^2}$

(4) $\frac{x_1 + x_2 - 2x_3}{10^6 Y^2}$

5) **Assertion (A) :** Gold chloride ($AuCl_3$) solution cannot be stored in a vessel made of Copper, iron, nickel, chromium and zinc.
Reason (R) : Gold has very high standard oxidation potential.

(1) Both assertion and reason are true but reason is not correct explanation of assertion

(2) Assertion is true but reason is false

(3) Assertion is false but reason is true

(4) Both assertion and reason are true and reason is a correct explanation of assertion

6)

If K_1, K_2, K_3 are equilibrium constant for formation of AD, AD_2, AD_3 respectively as follows
$A+D \rightleftharpoons AD$, $AD+D \rightleftharpoons AD_2$,
$AD_2 + D \rightleftharpoons AD_3$. Then equilibrium constant 'K' for $A + 3D \rightleftharpoons AD_3$ is related as

(1) $K_1 + K_2 + K_3 = K$

(2) $\log K_1 + \log K_2 + \log K_3 = \log K$

(3) $K_1 + K_2 = K_3 + K$

(4) $\log K_1 + \log K_2 = \log K_3 + \log K$

7) Which of the following mixture has maximum pH :-

(1) 200 mL, $\frac{M}{10}$ H_2SO_4 + 800 mL, $\frac{M}{10}$ NaOH

(2) 800 mL, $\frac{M}{10}$ HCl + 200 mL, $\frac{M}{10}$ $Ba(OH)_2$

(3) 500 mL, $\frac{M}{10}$ H_2SO_4 + 500 mL, $\frac{M}{10}$ $Ba(OH)_2$

(4) 250 mL, $\frac{M}{10}$ HNO_3 + 750 mL, $\frac{M}{10}$ KOH

8) Match the column :-

Column - I (Solute)		**Column-II (van't Hoff factor, i)**	
(A)	$AlCl_3$ if $\alpha = 0.8$	(P)	i = 3.4
(B)	$BaCl_2$ if $\alpha = 0.9$	(Q)	i = 2.8
(C)	CH_3COOH dissolved in benzene	(R)	i = 3.8
(D)	$K_4[Fe(CN)_6]$, if $\alpha = 0.7$	(S)	i = 0.5

(1) A→P, B→Q, C→R, D→S

(2) A→Q, B→P, C→S, D→R

(3) A→P, B→Q, C→S, D→R

(4) A→R, B→S, C→P, D→Q

9)

How many moles of sucrose should be dissolved in 500 gms of water as to get a solution which has a difference of 103.57°C between boiling point and freezing point :-
(K_f = 1.86 K kg mol^{-1}, K_b = 0.52 K kg mol^{-1})

(1) 0.5 moles

(2) 0.9 moles

(3) 0.75 moles

(4) 0.65 moles

10) In complete combustion of 0.1 mole of a hydrocarbon C_xH_y , 0.1 mole CO_2 & 0.2 mole H_2O is produced. Molecular formula of hydrocarbon is :

(1) CH_4

(2) C_2H_4

(3) C_2H_6

(4) C_4H_8

11) $K_2\underline{Mn}O_4$ = +7
$Ca\underline{O}_2$ = –1
$H_2\underline{S}_2O_7$ = +7
$Fe_3\underline{O}_4$ = $+\frac{8}{3}$
$\underline{Cr}_2O_7^{2-}$ = +6
$\underline{N}O_3^-$ = +5
The number of species in which oxidation number of underlined atom is correct ?

(1) 3

(2) 2

(3) 1

(4) 4

12) The enthalpy of combustion of benzene from the following data will be :- (i) $6C(s) + 3H_2(g) \rightarrow C_6H_6(\ell)$; $\Delta H = +45.9$ kJ (ii) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\square)$; $\Delta H = -285.9$ kJ (iii) $C(s) + O_2(g) \rightarrow CO_2(g)$; $\Delta H = -393.5$ kJ

(1) 3172.8 kJ

(2) –1549.2 kJ

(3) –3172.8 kJ

(4) –3264.6 kJ

13) For the reaction taking place at certain temperature $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$. If equilibrium pressure is 3X bar then Δ_rG^0 would be.

(1) –RT ln 9 – 3RT ln X
(2) RT ln 4 – 3RT ln X
(3) – 3RT ln X
(4) – RT ln 4 – 3RT ln X

14) One mole of an ideal gas expand reversibly and isothermaly from 3L volume to 30L volume at temperature TK. The work done by gas is 8.31 kJ then calculate temperature T.

(1) 200 K
(2) 500 K
(3) 868.4 K
(4) 434.2 K

15) For which of the following reactions $K_p < K_c$:-

(1) $N_2O_3(g) \rightleftharpoons NO_2(g) + NO(g)$
(2) $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$
(3) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(4) $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

16) Which of the following is gaseous element of s-block ?

(1) Neon
(2) Nitrogen
(3) Hydrogen
(4) All of these

17) Match the column

	Order		**Property**
(A)	In < T□	(P)	Ionisation energy
(B)	F > C□	(Q)	Electronegativity
(C)	Na^+ > Ne	(R)	Isoelectronic pair
(D)	A□ > Ga	(S)	Size
		(T)	Same group element

(1) A→P,Q,S,T; B→P,Q,T; C→P,Q,R; D→S,T
(2) A→Q,S,T; B→Q,T; C→P,Q, D→S
(3) A→P,Q; B→P,Q,R,S; C→Q,R,S; D→S,T

(4) A→P,Q,S,T; B→P,Q,T; C→Q,R; D→T

18) Which of the following statement is not correct ?

(1) In the σ bond formation atomic orbitals overlap in such a way that their axis and internuclear axis remains perpendicular to each other

(2) The hybridised orbitals are always equivalent in energy and shape

(3) In M.O.T. total energy of two molecular orbitals remains the same as that of two original atomic orbitals

(4) The magnitude of H-bonding depends on the physical state of the compound

19) Identify the **incorrect** statement-

(1) When PEt_3 and $AsPh_3$ acts as ligands can form dπ-dπ bond with transition metals

(2) The N-N single bond is as strong as the P-P single bond.

(3) Nitrogen has unique ability to form pπ-pπ multiple bonds with nitrogen, carbon and oxygen

(4) Nitrogen cannot form dπ-pπ bond as other heavier elements of its group.

20) Among given molecules find which is/are polar, non-planar and having 1 lone pair on central atom ?
$XeF_5^+, XeF_5^-, SF_4, SnCl_2, ICl_3, SO_2$

(1) SF_4, ICl_3

(2) SO_2, XeF_5^-

(3) $XeF_5^+, SnCl_2$

(4) SF_4, XeF_5^+

21) Dipole-Dipole interaction exist between which pair :-

(1) KCl and H_2O

(2) C_6H_6 and C_2H_5OH

(3) C_6H_6 and CCl_4

(4) Acetone & acetonitrile

22) Which has maximum bond length among the following ?

(1) O – H

(2) C = O

(3) C = C

(4) C ≡ C

23) Consider the following complex $[Cr(NH_3)_2Cl_2(C_2O_4)]^-$ what is
(i) oxidation number of metal ion
(ii) Co-ordination number of metal ion
(iii) charge on complex if all ligands were chloride ions,
Choose the correct set.

	(i)	(ii)	(iii)
A	–1	6	–3
B	+3	6	–1
C	+3	6	–3
D	–1	5	–3

(1) A

(2) B

(3) C

(4) D

24) For complex ion, dichlorido bis-(ethylenediamine) cobalt (III) ion select the correct statement :-

(1) 3 isomers ; out of which 2 are optically active.

(2) 3 isomers ; all are optically active.

(3) 3 isomers ; all are optically inactive

(4) Only one optically active isomer & 3GI

25) Which of the following statement is incorrect.

(1) Greater the formation constant (K_f) of a complex ion, greater is its stability.

(2) Greater the charge on C.M.I. greater is the stability of complex.

(3) Greater is the electron pair donating ability of the ligand lesser is the stability of complex.

(4) Chelate complexes have high stability.

26) What is the oxidation state of Xe in compound, which is discovered by Neil Bartlett.

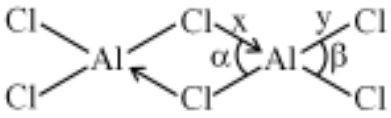
(1) +1

(2) +6

(3) +4

(4) None of these

27) Identify correct statement :



(i) $\alpha > \beta$, bond angle.

(ii) x > y, bond length

(iii) Hybridisation state of Al is sp^3

(iv) Octet of Al is complete so it is $e^{\ominus}$ deficient.

(1) (i) and (ii)

(2) (ii) and (iii)

(3) (iii) and (iv)

(4) (i) and (iv)

28) How many oxide are mixed oxide in given oxides
Mn_2O_7, Pb_3O_4, Bi_2O_3, Co_3O_4, Fe_3O_4, Mn_3O_4, Cr_2O_3

(1) 4
(2) 5
(3) 3
(4) 2

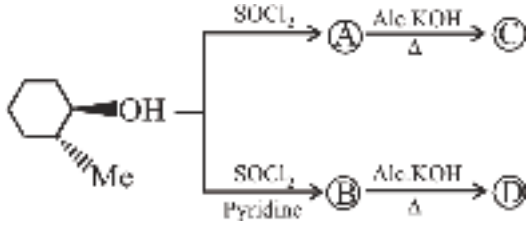
29) Incorrect statement for lanthanides is :

(1) They liberate hydrogen gas with dilute HCl.
(2) They form carbides with carbon on heating.
(3) Their hydroxides are basic.
(4) Ce^{+4} is good reducing agents.

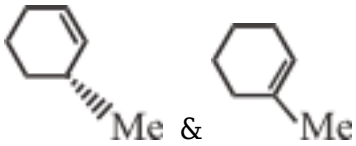
30) Which set of anions gives same colour gas with dilute acid and concentrated acid respectively.

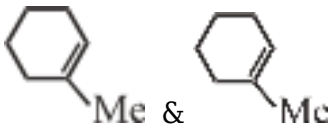
(1) Br^-, NO_3^-
(2) NO_2^- , I^-
(3) I^-, NO_2^-
(4) NO_2^- , Br^-

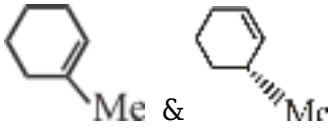
31)

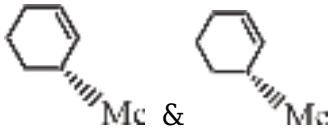


Major product C and D will be respectively:

(1) 

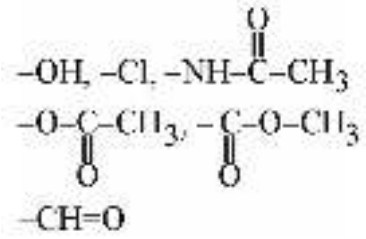
(2) 

(3) 

(4) 

32)

How many of following are ortho-para directing in electrophilic substitution reaction

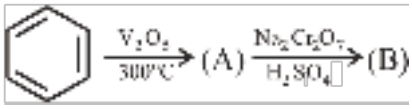


(1) 4

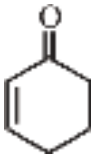
(2) 6

(3) 5

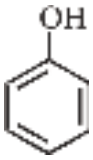
(4) 3

33) 

Product (B) is :-

(1) 

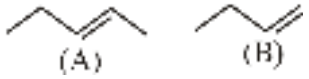
(2) 

(3) 

(4) 

34) Which are correct options :-

(i)  Stability B > A

(ii)  Heat of combustion A > B

(iii)  Heat of hydrogenation A > B

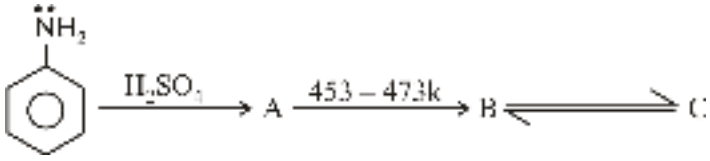
(iv)  Bond length of C = C B > A

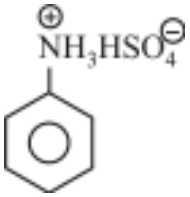
(1) Only i, ii

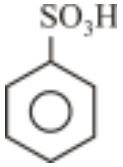
(2) Only i, ii, iii

(3) Only i, iii, iv

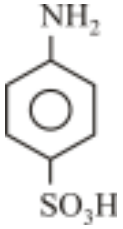
(4) i, ii, iii, iv

35) 

(a) compound (A) is 

(b) compound (B) is 

(c) compound (C) is zwitter ion

(d) compound (B) is 

which of the above statement is/are correct ?

(1) a, b and c

(2) a, c, d

(3) b, c, d

(4) a, b, c ,d

36) Acetic anhydride is obtained by the reaction of :-

(1) Sodium and acetic acid

(2) Ammonia and acetic acid

(3) P_2O_5 and acetic acid

(4) Ethanol and acetic acid

37) Which of the following reaction is incorrect ?

(1) $CH_3CH_2OH \xrightarrow[2.H^+/H_2O]{1.\text{alkaline } KMnO_4} CH_3-COOH$

(2) $CH_3CONH_2 \xrightarrow[\Delta]{H_3\overset{\oplus}{O}} CH_3-COOH$

(3) $CH_3CN \xrightarrow{\text{Complete Hydrolysis}} CH_3-COOH$

(4) $CH_3COCl \xrightarrow{H_2O} CH_3CHO$

38) **Assertion (A) :** All naturally occurring α-amino acids except glycine are optically active.

Reason (R) : Most naturally occurring amino acids have L-configuration.

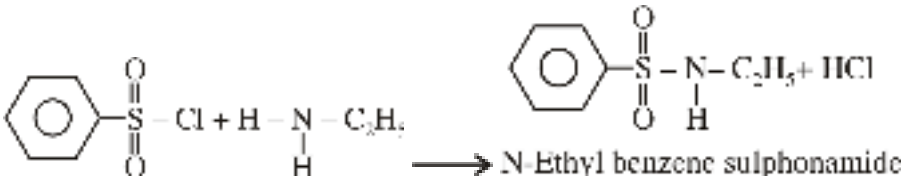
(1) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

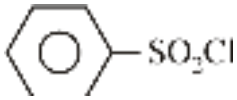
(3) Assertion (A) is true and Reason (R) is false

(4) Assertion (A) is false and Reason (R) is also false

39) Identify true (T) and false (F) statements in the given following statements respectively :-
(A) Primary aliphatic amines react with nitrous acid to form aliphatic diazonium salt which being unstable

(B) 

In above reaction, N-Ethyl benzene sulphonamide soluble in alkali

(C)  is called as Hinsberg reagent

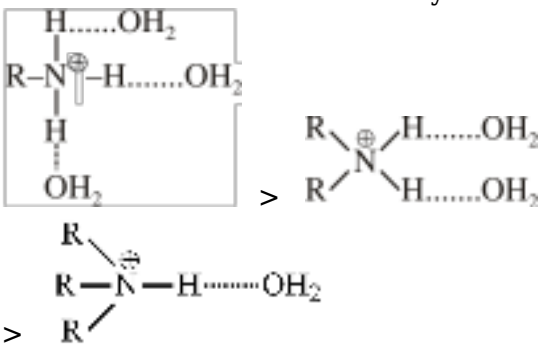
(1) T, T, T

(2) T, T, F

(3) T, F, F

(4) F, F, F

40) Given below are two statement :
Statement I : Order of basic strength in case methyl substituted amine in aqueous solution $(CH_3)_2$ NH > CH_3NH_2 > $(CH_3)_3$ N > NH_3
Statement II : Order of stability of ions by solvation



(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct and Statement II is incorrect.

(4) Statement I is incorrect and Statement II is correct.

41) Which of the following statement is not correct :

(1) F_2 > Cl_2 > Br_2 > I_2
(Reactivity of halogens towards alkane)

(2) $3° > 2° > 1°$ H
(Rate of Replacement of hydrogen by halogen of alkane)

(3) Iodonation is very slow and reversible reaction

(4) Oxidising agent like HIO_3, HNO_3 are used during fluorination.

42) Which of the following general formula is not correctly matched ?

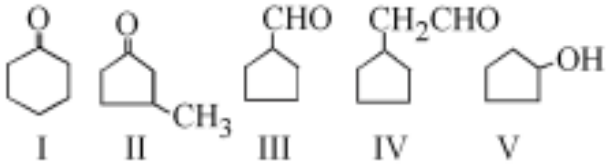
(1) Alkadiene C_nH_{2n-2}

(2) Carboxylic acid $C_nH_{2n}O_2$

(3) Carboxylic anhydride $C_nH_{2n}O_3$

(4) Ether $C_nH_{2n+2}O$

43) Which of the following are isomers



(1) I, II Only

(2) III, IV Only

(3) I, II, III Only

(4) All of the above

44)

Column I (Process)			Column II (Sepration)
A.	Sublimation	P	Impure compound heated after that cooling the solution pure compound crystallize out and is remove by filtration
B.	Crystallisation	Q	Separation in which one compound sublimate another not sublimate
C.	Distillation	R	Volatile liquid from non-volatile impurities
D.	Differential Extraction	S	When an organic compound is present in an aqueous medium it separated by shaking it with an organic solvent in which it more soluble than water

	A	B	C	D
(1)	P	Q	R	S
(2)	S	R	Q	P
(3)	Q	P	S	R

(4)	Q	P	R	S

(1) 1

(2) 2

(3) 3

(4) 4

45) A carbonyl compound 'X' on reaction with NH_2OH forms compound that can show geometrical isomerism and 'X' give reddish brown precipitate on reaction with Fehling's solution. Then X can be :-

(1) Formaldehyde

(2) Acetaldehyde

(3) Benzaldehyde

(4) Acetone

BIOLOGY

1) Match the column I with column II for wheat classification and select the correct option using the codes given below.

	Column-I		**Column-II**
(a)	Division	(i)	Poales
(b)	Order	(ii)	Poaceae
(c)	Family	(iii)	Monocotyledonae
(d)	Class	(iv)	Angiospermae

(1) a-ii, b-iii, c-iv, d-i

(2) a-iii, b-iv, c-i, d-ii

(3) a-iv, b-i, c-iii, d-ii

(4) a-iv, b-i, c-ii, d-iii

2) Which of the following is not correctly matched ?

	Genus Name	**Character**	**Class**
(1)	*Tricoderma*	Sexual stage is absent	Deuteromycetes
(2)	*Claviceps*	Endogenous ascospores	Ascomycetes
(3)	*Ustilago*	Septate mycelium	Basidiomycetes

(4)	*Puccinia*	Aseptate and coenocytic mycelium	Phycomycetes

(1) 1

(2) 2

(3) 3

(4) 4

3) Match the column -

	Column-I		**Column-II**
(i)	Virus	(a)	Composite organism
(ii)	Viroids	(b)	Protein enclosing nucleic acid
(iii)	Prions	(c)	Abnormally folded protein
(iv)	Lichen	(d)	Low molecular RNA.

(1) i-b, ii-c, iii-d, iv-a

(2) i-b, ii-d, iii-a, iv-c

(3) i-b, ii-d, iii-c, iv-a

(4) i-c, ii-b, iii-d, iv-a

4) Match the column I and Column II :

	Column-I		**Column-II**
(P)	*Gonyaulax*	(i)	Soap box
(Q)	Slime moulds	(ii)	Red tides
(R)	Chrysophytes	(iii)	Released toxins for marine animals
		(iv)	Spread over several feet
		(v)	Spores survive for many years
		(vi)	Golden algae
		(vii)	Diatomaceous earth

(1) P → vi, vii; Q → iv, v ; R → i, ii, iii

(2) P → iv, v; Q → i, ii, iii ; R → vi, vii

(3) P → ii, iii; Q → iv, v ; R → i, vi, vii

(4) P → iv, v ; Q → vi, vii ; R → i, ii, iii

5) Read the column-I, II and III and identify their correct match –

	Column-I		**Column-II**		**Column-III**
(A)	Chlorophyceae	(P)	Protonema	(i)	Pollination
(B)	Bryophyta	(Q)	Oogonium	(ii)	Mosses
(C)	Pteridophyta	(R)	Ovule	(iii)	Zygotic meiosis
(D)	Gymnosperm	(S)	Prothallus	(iv)	*Psilotum*

(1) (A)-(P)-(i), (B)-(Q)-(ii), (C)-(R)-(iii), (D)-(S)-(iv)

(2) (A)-(S)-(iv), (B)-(R)-(iii), (C)-(Q)-(ii), (D)-(P)-(i)

(3) (A)-(R)-(i), (B)-(S)-(iv), (C)-(P)-(ii), (D)-(Q)-(iii)

(4) (A)-(Q)-(iii), (B)-(P)-(ii), (C)-(S)-(iv), (D)-(R)-(i)

6) **Statement-I** :- In *Calotropis* valvate aestivation is present.
Statement-II :- In *Cassia* and gulmohar the aestivation is vexillary.
In the light of above statements choose the correct answer from the following option.

(1) Statement I is correct and II is incorrect

(2) Both statement are correct

(3) Statement I is incorrect and II is correct

(4) Both statement are incorrect

7) In cymose type of inflorescence :
(A) Main axis terminate into flower.
(B) Flowers borne in basipetal order.
(C) Main axis do not terminate into flower.
(D) Flowers borne in acropetal order.
Find out the correct option :

(1) A and B only

(2) C and D only

(3) A and C only

(4) B and C only

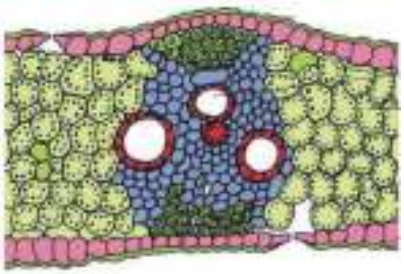
8) Identify the incorrect statement.

(1) All tissues on the inner side of the endodermis such a pericycle, vascular bundles, and pith constitute the stele.

(2) Pith is large and well developed in monocot root.

(3) Monocot roots do not undergo any secondary growth.

(4) The monocot stem has, parenchymatous hypodermis.

9)  Identify the given diagram and select the correct option with respect to given diagram ?

(1) Hypostomatic leaf

(2) Amphistomatic leaf

(3) Found in sunflower plant

(4) Both 2 and 3

10) How many of the following statements are **correct** ?
(a) The prokaryotic flagella are structurally different from the eukaryotic flagella
(b) Middle lamella layer mainly consist of calcium carbonate
(c) The quasi-fluid nature of lipids enables lateral movement of proteins within the overall bilayer
(d) Polar molecules can pass through the non-polar lipid bilayer easily.

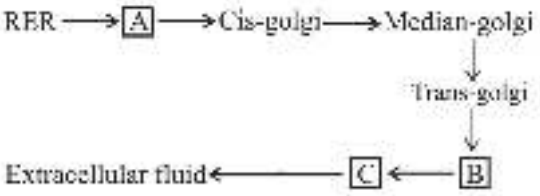
(1) Two

(2) Three

(3) Four

(4) One

11) Choose the correct for given flow chart:



	A	B	C
(1)	ER vesicle	Mitochondrion	Secretory vesicle
(2)	Secretory vesicle	Nucleus	Plasma membrane
(3)	ER vesicle	Secretory vesicle	Plasma membrane
(4)	Secretory vesicle	Plasma membrane	ER vesicle

(1) 1

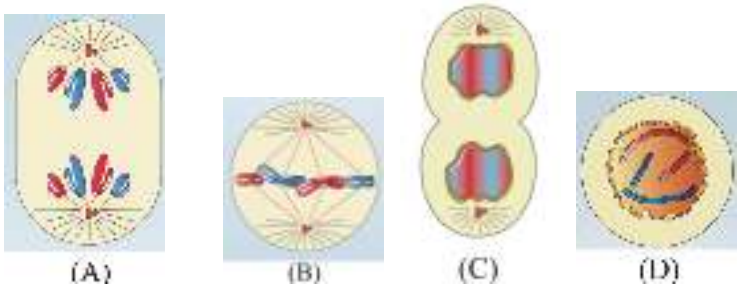
(2) 2

(3) 3

(4) 4

12)

The four figures (A, B, C and D) given below represent four different stages of mitotic cell division. Which one of these is correctly identified in the options given, alongwith its correct description and key events ?



		Stage	Description	Key event
(1)	D	Prophase	Followed by G_2 Phase	Bivalent Chromosomes align on the equatorial plate
(2)	B	Metaphase	Condensation of chromosomes completed	Daughter chromosomes move to opposite poles
(3)	C	Telophase	Longer and highly complex	Homologous chromosomes separate
(4)	A	Anaphase	Centromeres are towards poles and arms trailing behind	Centromere split and chromatids separate

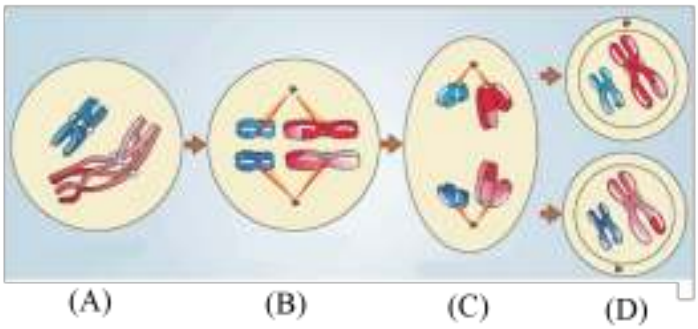
(1) 1

(2) 2

(3) 3

(4) 4

13)



Choose the **correct** option for given diagram :-

(1) **A**-Crossing over

(2) **B**-Segregation of homologous chromosomes

(3) **C**-Pairs of homologous chromosomes are align on the equatorial plate

(4) **D**-Nucleolus disappear

14) Which of the following statement is **not** true ?

(1) Cell plate represents the middle lamella between the walls of two adjacent cells

(2) At the time of cytokinesis organelles like mitochondria and plastids, get distributed between the daughter cells

(3) Cytokinesis in plant cell is centrifugal and takes place by cell plate formation while in animal cells by furrowing or cleavage and it is centripetal

(4) M–phase is last for 95% of the duration of cell cycle

15) With regard to concept of activation energy, identify the **wrong** statements :
(A) If product is at a lower energy level than substrate, the reaction is an endothermic reaction.
(B) The difference in average energy content of substrate from that of transition state is called activation energy.
(C) The pathway of transformation of substrate into product must go through the transition state.
(D) There could be only one altered structural state between the stable substrate and the product.
(E) Enzymes eventually bring down the activation energy barrier.
Choose the correct answer from the options given below :

(1) A, C and D only

(2) A, B, C and D only

(3) A and D only

(4) B, C and D only

16) Identify the correct pair :-

Biomolecule		**Function/Nature**	
(1)	Collagen	(a)	Fights infectious agents
(2)	GLUT-4	(b)	Hormone
(3)	Inulin	(c)	Polymer of fructose
(4)	Trypsin	(d)	Carbohydrate

(1) 1

(2) 2

(3) 3

(4) 4

17) Which of the following statements is **correct** ?

(1) Ricin and morphine are toxins

(2) Concanvalin 'A' is a type of alkaloid

(3) Mono and diterpenes are considered as secondary metabolites

(4) Vinblastin is a type of essential oil

18) Select the **correct** statements :
(a) Plasticity is the ability of plants to always follow same pathway in response to environment or phases of life to form different kinds of structures.
(b) Cotton, Coriander and larkspur are heterophyllous plants.
(c) The growth, differentiation and development are very closely related events in the life of a plant.
(d) Development in plants is not under the control of intrinsic or extrinsic factors.

(1) a, b and c

(2) b and d

(3) b and c

(4) b, c and d

19) Which of the following statements are **incorrect**.
A. The gaseous PGR is ABA.
B. Gibberellins induce internode elongation just prior to flowering in rosette plants (Beet root and cabbage)
C. Removal of root tips can overcome apical dominance
D. Ethylene is also known as stress hormone.

(1) A and B only

(2) A, B and C only

(3) A, C and D only

(4) A and D only

20) Select the **correct** statement :-

(1) Present concentration of atmospheric CO_2 is limiting for C_4 plants

(2) Tomatoes and bell pepper are allowed to grow in CO_2 enriched atmosphere that leads to higher yields.

(3) Light is major limiting factor for photosynthesis

(4) CO_2 is limiting factor in cloudy days.

21) **Statement –I** :- When only PSI is functional, the electron is circulated within the photosystem and the phosphorylation occurs due to cyclic flow of electrons
Statement –II :- Each photosystem has all the pigments (except one molecule of chlorophyll–a) forming a light harvesting system also called antennae.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

22) Which of the following is/are **correct** about cellular respiration ?
(A) The respiratory pathway is an amphibolic pathway as it involves both anabolism and catabolism.
(B) In eukaryotes the entire cellular respiration takes place in cytoplasm.
(C) Fermentation takes place under aerobic condition in germinating seeds.
(D) Metabolic fate of pyruvate also depends on the cellular need.
(E) Water is formed during respiration as a result of O_2 accepting electrons and getting oxidised.
Choose the correct answer from the options given below:

(1) A, B, C, D only

(2) A and D only

(3) A, C, D, E only

(4) A only

23) Link reaction/oxidative decarboxylation in respiration occurs in which part of cell ?

(1) Cytoplasm
(2) Inner membrane of mitochondrion
(3) Mitochondrial matrix
(4) outer membrane of mitochondrion

24) **Assertion :** The functional megaspore undergoes three successive mitotic division to form on 8-nucleated embryo sac, but only seven cells are formed because two polar nuclei remain in a single large central cell.
Reason : The filiform apparatus is present in egg cell.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is true but **Reason** is false
(4) **Assertion** is false but **Reason** is true

25) Following are the steps involved in the process of embryo development in dicot ?
(A) Heart-shaped embryo
(B) Mature embryo
(C) Globular embryo
(D) Zygote
Choose the correct sequence of step of embryo development from the option given below :-

(1) D → A → C → B
(2) C → B → D → A
(3) B → C → A → D
(4) A → D → B → C

26) The given plant is pollinated by :-



(1) Wind
(2) Water
(3) Insect
(4) Bird

27) When F_1 tall plants were self pollinated, which of the following was not observed by Mendel in F_2-generation ?

(1) The tall and dwarf traits were identical to their parental type.

(2) Parental traits were not blended in F_2 plants

(3) $3/4^{th}$ of the F_2 plants were pure tall.

(4) $1/4^{th}$ of the F_2 plants were dwarf.

28) In Morgan's experiment, what was the percentage of recombinants in *Drosophila* w.r.t body colour and eye colour ?

(1) 37.2%

(2) 98.7%

(3) 1.3%

(4) 62.8%

29) Blood group of a man is 'O' and that of his wife is AB then their children have only blood group:-

(1) O, AB

(2) O, A, AB

(3) O, B, A

(4) A, B

30) The distance between genes W, X, Y and Z on a chromosome are as follows. W – Y = 18 units

W – X = 26 units
W – Z = 40 units
X – Y = 8 units
X – Z = 14 units The sequence of these genes would be :-

(1) W, X, Y, Z

(2) X, Y, W, Z

(3) W, Y, X, Z

(4) Y, W, X, Z

31) Which of the following condition correctly describe the manner of determining the sex in the given example ?

(1) XX type of sex chromosome determine the male sex in grasshoppers.

(2) XO conditions in human as found in Turner's syndrome, determine male sex

(3) Homozygous sex chromosome (XX) produce female in *Drosophila.*

(4) Homozygous sex chromosome ZZ determine female sex in birds.

32) Given below is a list of some pairs of contrasting traits in pea plant with some blanks:

Character	**Dominant trait**	**Recessive trait**
Seed shape	Round	I

Flower Colour	II	White
Pod Shape	Full	III
IV	Axial	V

Choose the option which correctly fill up the above blanks:

(1) I (wrinkled), II (violet), III(Constricted), IV (flower position), V (terminal)

(2) I (wrinkled), II (Purple), III (constricted), IV (flower position), V (Terminal)

(3) I (wrinkled), II (violet), III (Round), IV (flower position), V (Terminal)

(4) I (wrinkled), II (green), III (constricted), V (flower position), VI (Terminal)

33) Read the following statements :

(i) Eukaryotic mRNA is polycistronic whereas prokaryotic mRNA is monocistronic

(ii) Prokaryotic mRNA is capless and tail-less whereas eukaryotic mRNA is with cap and tail

(iii) The precursor of prokaryotic mRNA is hnRNA

(iv) In replication only a segment of DNA and only one of the strand is copied into RNA

(v) In eukaryotes, transcription occurs in nucleus and translation in cytoplasm

Which of the above statements are correct ?

(1) (ii), (iii) and (v)

(2) (iii), (iv) and (v)

(3) (ii) and (v)

(4) (iii) and (iv)

34) **Assertion (A) :** In eukaryotes, the primary transcript (hnRNA) must undergo splicing, capping, and polyadenylation before becoming functional mRNA, because raw transcripts cannot directly participate in translation.

Reason (R) : These post-transcriptional modifications protect RNA from exonuclease degradation, enhance transport from nucleus to cytoplasm, and promote efficient ribosome recognition during protein synthesis.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

35) **Assertion :** In the first phase of translation amino acids itself are activated in the presence of ATP.

Reason : If two such charged tRNAs are brought close enough then the formation of peptide bond between them would be favoured energetically.

(1) Both Assertion and Reason are correct Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are incorrect.

(3) Both Assertion and Reason are correct Reason is not correct explanation of Assertion.

(4) Assertion is wrong Reason is correct.

36) Match the following column –

Column-I		Column-II	
(A)	UUU	(i)	Start codon
(B)	GUG	(ii)	Stop codon
(C)	GAG	(iii)	Code for valine
(D)	AUG	(iv)	Code for glutamic acid
(E)	UGA	(v)	Code for phenyl alanine

	(A)	(B)	(C)	(D)	(E)
(1)	(v)	(iv)	(iii)	(i)	(ii)
(2)	(v)	(iii)	(iv)	(i)	(ii)
(3)	(v)	(iv)	(iii)	(ii)	(i)
(4)	(v)	(i)	(iii)	(iv)	(ii)

(1) 1
(2) 2
(3) 3
(4) 4

37) Choose incorrect statement ?

(1) Heterochromatin is transcriptionally inactive.
(2) NHC protein is required for packaging of chromatin at higher level.
(3) Histone octamer consists of five type of histone protein.
(4) Histone rich with lysine & arginine amino acids.

38) Polymorphisms play very important role in :-

(1) DNA finger printing
(2) Evolution
(3) Genetic mapping
(4) All of the above

39) The unequivocal proof of DNA as the genetic material came from the studies of virus, was first proposed by

(1) Hershey and chase
(2) Avery, MacLeod and McCarthy
(3) Wilkins and Franklin
(4) Fredrick Griffith

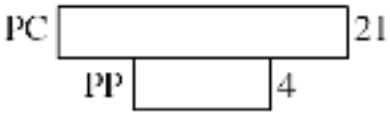
40) Which one of the following is defining feature of parasitism-

(1) The life cycle of parasite is always complex

(2) Adhesive organs are absent in parasite

(3) Parasite lost unnecessary sense organs

(4) Parasites have well developed digestive system



41) Recognise the figure and find out the given figure represents:

(1) Pyramid of number in tree ecosystem

(2) Pyramid of number of grassland ecosystem

(3) Pyramid of Biomass in aquatic ecosystem

(4) Pyramid of Biomass in tree ecosystem

42) One out of the following is not responsible for biodiversity loss :

(1) Alien species Invasion

(2) Co-extinction

(3) Ex-situ conservation

(4) Deforestation

43) Read the following column-I & column-II and identify correct match :–

	Column-I		**Column-II**
(A)	Interspecific competition	(i)	Potent force in organic evolution
(B)	Intraspecific competition	(ii)	Gause
(C)	Predator	(iii)	David tileman
(D)	Competition exclusion principle	(iv)	Most acute struggle
		(v)	Prudent

(1) (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)

(2) (A)-(ii), (B)-(iii), (C)-(v), (D)-(i)

(3) (A)-(i), (B)-(iv), (C)-(v), (D)-(ii)

(4) (A)-(iv), (B)-(iii), (C)-(i), (D)-(v)

44) **Assertion :** The annual net primary productivity of the whole biosphere is approximately 170 million tons (dry weight) of organic matter.
Reason : Of this despite occupying about 70 percent of the surface. The productivity of ocean are only 55 millions tons.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) Both **Assertion** and **Reason** are false.

45) Read the following statements :-
(I) Recent studies do not support gross generalisations about competition as suggested in "Gause's competitive exclusion principle".
(II) The monarch butterfly is highly distasteful to its predator (bird) because of a special chemical present in body. Intrestingly, the butterfly acquires this chemical during its caterpillar stage by feeding on poisonous weed.
(III) In primary succession in water, the pioneer community is rooted hydrophytes.
(IV) Species diversity, dominance, stratification are structural characteristics of community.
Which of the above statements are correct ?

(1) I, II, IV

(2) II, IV

(3) III

(4) I, III, IV

46) How many characters are true for pleurobrachia ?
Sea walnuts, ciliated comb plates, exclusively marine, triploblastic, bilateral symmetry, flame cells.

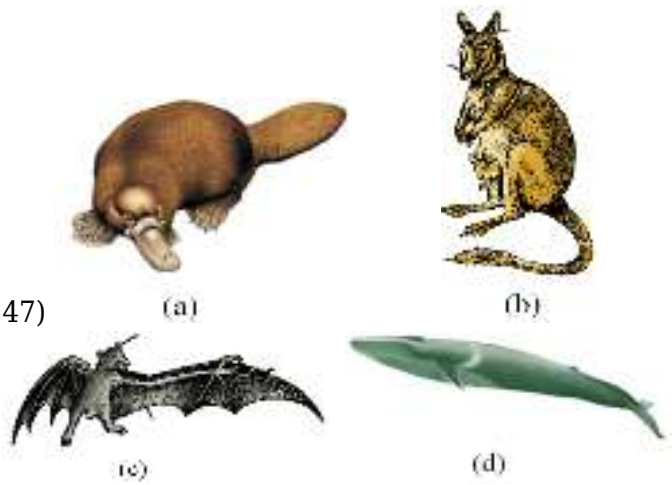
(1) Four

(2) Three

(3) Two

(4) Five

47)



Identified the figure and choose the correct option :

(1) a-pteropus, b-ornithorynchus, c-macropus, d-balaenoptera

(2) a-ornithorynchus, b-pteropus, c-macropus, d-balaenoptera

(3) a-ornithorynchus, b-macropus, c-pteropus, d-balaenoptera

(4) a-ornithorynchus, b-balaenoptera, c-macropus, d-pteropus

48) Which of the following are related with phylum Annelida ?

(1) Notochord

(2) Radial - symmetry

(3) Diploblastic

(4) Presence of metameres.

49) Fill up with the suitable words
Cnidarians exhibit two basic body forms called __a__ and __b__. The former is __c__ and __d__ form whereas, the latter is __e__ and __f__.

	a	b	c	d	e	f
1	polyp	medusa	sessile	cylindrical	Umbrella shaped	Free swimming
2	polyp	medusa	Umbrella shaped	cylindrical	Sessile	Free swimming
3	medusa	polyp	Umbrella shaped	cylindrical	Sessile	Free swimming
4	medusa	polyp	sessile	cylindrical	Umbrella shaped	Free swimming

(1) 1

(2) 2

(3) 3

(4) 4



50) Which of the following statement / character is incorrect about the diagram given above ?

(1) Posses lateral appendages , parapodia which help in reproduction.

(2) Body surface distinctly marked out into segments.

(3) Circulatory system is closed.

(4) Nephridia help in excretion and osmoregulation.

51) Choose the incorrect statements about Aves.
I. Endoskeleton is fully ossified and the long bones are hollow with air cavities.
II. The digestive tract of birds have additional chambers, the crop and gizzard.
III. In birds heart is three chambered.

IV. Birds cannot maintain constant body temperature.
V. Other than tail, oil glands are also present on the skin.

(1) III, IV and V only
(2) I, II and III only
(3) I and II only
(4) II and III only

52) _____ forms a diffusion boundary in our body :-

(1) Compound epithelium
(2) Simple columnar epithelium
(3) Simple squamous epithelium
(4) Simple cuboidal epithelium

53) The zoological name of cockroach is

(1) *Periplaneta americana*
(2) *Pheretima posthuma*
(3) *Rana tigrina*
(4) *Naja naja*

54) Which of the following statement is correct regarding frog ?

(1) Ventricle opens into a sac like conus arteriosus on the dorsal side of heart.
(2) Sinus venosus receive deoxygenated blood from right atrium.
(3) Bidder's canal is present in male frog kidney.
(4) Copulatory pad is present on first digit of hind limb in male frog.

55) Which of the following is correctly match regarding frog ?

(1) Capulatory pad - Helps in insemination.
(2) Mesorchium - Single fold of peritoneum.
(3) Hind brain - Cerebrum and medulla oblongata.
(4) RBC - Biconvex and nucleated.

56) **Assertion** : ECG obtained from different individuals have roughly the same shape for a given lead configuration.
Reason : ECG have a great clinical significane.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(3) **Assertion** is false but **Reason** is true.
(4) **Assertion** is true but **Reason** is false.

57) In a standard ECG which of the following positions of lead is not correct ?

(1) Right wrist

(2) Left wrist

(3) Right ankle

(4) Left ankle

58) Blood pressure in human arteries is maximum during :

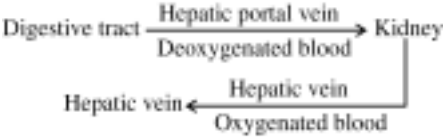
(1) Systole of both atria

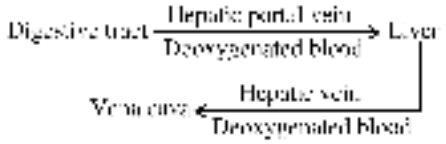
(2) Systole of both ventricles

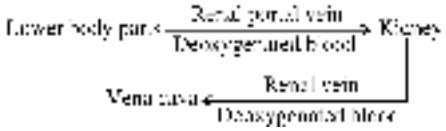
(3) Diastole of both ventricles

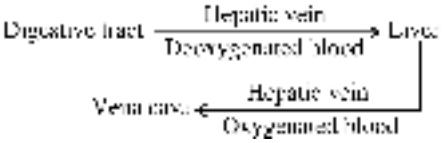
(4) Diastole of both atria

59) Choose correct for hepatic portal circulation.

(1) 

(2) 

(3) 

(4) 

60) **Statement-I :-** Nearly 20-25% of CO_2 is transported in the form of bicarbonate in plasma.
Statement-II :- The role of O_2 in the regulation of respiration is quite significant.
Choose the appropriate option :-

(1) Statement-I and statement-II both are correct.

(2) Statement-I and statement-II both are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

61) Bohr effect enhances oxygen delivery at the tissue level because :-

(1) Low pH increases O_2 binding to Hb

(2) CO_2 stabilizes oxyhaemoglobin

(3) Increased H^+ reduces O_2 binding affinity to Hb

(4) Hb is denatured by H_2CO_3

62) **Statement-I :-** In majority of nephrons, the loop of Henle is too short and extend only very little

into the medulla.
Statement-II :- Glomerulus is a tuft of capillaries formed by afferent arteriole.

(1) Both statement-I & statement-II are true.
(2) Statement-I is true but statement-II is false.
(3) Both statement-I & statement-II are false.
(4) Statement-I is false but statement-II is true.

63) Which of the following is incorrect ?

(1) ANF can cause vasodialation
(2) ANF acts as a check on the renin-angiotension mechanism
(3) ANF decrease the blood pressure
(4) ANF increase the blood pressure.

64) (A) 1100-1200 ml of blood is filtered by the kidneys per minute.
(B) A fall in GFR can activate the JG-cells to release renin.
(C) The ascending limb is permeable to water in Henle's loop.
(D) ADH facilitates water reabsorption from later parts of renal tubule.
(E) Henle's loop and vasa recta play role in counter current mechanism.

(1) A, B, C, & D are correct.
(2) A, B, D, & E are correct.
(3) A, C & E are correct.
(4) A, D & E are incorrect.

65) Given below two statements choose the correct statement :-
Statement-I : Myelinated nerve fibres and unmyelinated nerve fibres both are enclosed by Schwann cells but Schwann cells form a myelin sheath only in myelinated nerve fibred.
Statement-II : Myelinated and unmyelinated nerve fibred are found in PNS.

(1) Both Statements are correct.
(2) Both Statements are incorrect.
(3) Statement-I correct and Statement-II incorrect.
(4) Statement-I incorrect and Statement-II correct.

66) What is the general function of the sympathetic nervous system ?

(1) Promoting rest and digestion
(2) Preparing the body for "Fight or Flight".
(3) Regulating sleep-wake cycles.
(4) Controlling voluntary movements.

67) Nerve impulse is also termed as -

(1) Resting membrane potential
(2) Electric potential

(3) Action potential

(4) Graded potential

68) Which is not correct among following?

(1) Hormones from hypothalamus reach anterior pituitary through portal circulatory system.

(2) In males, FSH and androgens regulate spermatogenesis

(3) Acromegaly, always leads to death

(4) Neurohypophysis, store and release two hormones.

69) Which of the following performs dual functions as a primary sex organ as well as an endocrine gland ?

(1) Testis

(2) Pancreas

(3) Ovary

(4) 1 and 3 both

70) Male sexual behaviour is influenced by :-

(1) ACTH

(2) Oestrogen

(3) Testosterone

(4) Progesterone

71) The __A__ amino acid is the precursor for the synthesis of thyroxine hormone.
Correct option for A is :-

(1) Tryptophen

(2) Serotonin

(3) Tyrosin

(4) Phenyl alanin

72) Given below are two statements :
Statement-I : Small amounts of androgenic steroids are secreted by the adrenal cortex which play a role in the growth of axial hair, public hair and facial hair during puberty.
Statement-II : Aldosterone acts mainly at the renal tubules and stimulates the reabsorption of Na^+ and water and excretion of K^+ and phosphate ions and helps in the maintenance of electrolytes, body fluid volume, osmotic pressure and blood pressure.

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

73) Match the Column-I and Column-II

	Column-I		Column-II
(i)	Skeleton muscle	(a)	Striated, branched, involuntary
(ii)	Smooth muscle	(b)	Striated, voluntary
(iii)	Cardiac muscle	(c)	Non striated, involuntary

(1) i-(b), ii-(c), iii-(a)

(2) i-(c), ii-(b), iii-(a)

(3) i-(a), ii-(b), iii-(c)

(4) i-(b), ii-(a), iii-(c)

74) Some specialised cells in our body like macrophage and leucocytes shows which type of movement ?

(1) Muscular movement

(2) Cilliated movement

(3) Amoeboid movement

(4) Flagellar movement

75) Each coxal bone is formed by fusion of -

(1) Ilium, Ischium and Pubis

(2) Malleus, Incus and Stapes

(3) Thoracic, Lumbar and Cervical

(4) Tibia, Fibula, Patella

76) Match the following events with their resultant cells :

Column-1		Column-2	
A.	Meiosis-l	1.	Sperms
B.	Meiosis-II	2.	Spermatogonia
C.	Differentiation	3.	Ootid
D.	Mitosis	4.	First polar body

(1) A-4, B-1, C-3, D-2

(2) A-4, B-3, C-1, D-2

(3) A-4, B-1, C-2, D-3

(4) A-2, B-3, C-4, D-1

77) **Assertion :** Spermatogenesis starts at the age of puberty.
Reason : At the age of puberty there is a significant decrease in secretion of GnRH.

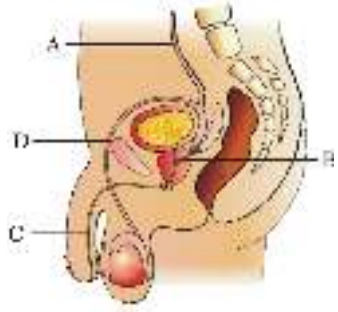
(1) Both **Assertion** and **Reason** false.

(2) **Assertion** is true and **Reason** is false.

(3) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(4) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

78) It is a diagrammatic sectional view of male reproductive system, In which identify ejaculatory



duct :-

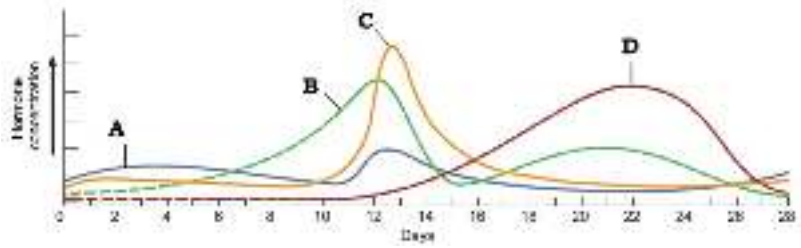
(1) A

(2) B

(3) C

(4) D

79) Identify the hormones A, B, C and D from the graph by observing their concentrations in blood during the menstrual cycle of a female:



	A	B	C	D
(1)	FSH	Progesterone	Estrogen	LH
(2)	FSH	Estrogen	LH	Progesterone
(3)	FSH	LH	Progesterone	Estrogen
(4)	LH	Estrogen	Progesterone	FSH

(1) 1

(2) 2

(3) 3

(4) 4

80)

How many of the following are used to treat allergy?
Histamine, steroids, adrenaline, Serotonin

(1) One

(2) Two

(3) Three

(4) Four

81) Match the immunity type with the example

	Immunity		Example
(A)	Active immunity	(i)	Antibodies
(B)	Passive immunity	(ii)	T-Lymphocyte
(C)	Humoral immunity	(iii)	Colostrum
(D)	Cell-mediated immunity	(iv)	Vaccination

(1) A-i, B-ii, C-iii, D-iv

(2) A-iv, B-iii, C-i, D-ii

(3) A-iv, B-iii, C-ii, D-i

(4) A-iii, B-ii, C-i, D-iv

82) Acquired immunity is pathogen specific. It is characterised by memory. This means when our body encounters a pathgen for the first time it produces a response called ?

(1) Anamnestic response

(2) Secondary immune response

(3) Primary immune response

(4) Exaggerated response

83) **Assertion (A) :** Helminthic infections such a ascariasis generally lead to abdominal pain, internal bleeding, muscular weakness and anemia inaffected individuals.
Reason (R) : These symptoms occurs because adult worm of ascaris lumbricoides absorb nutritents directly from the host intestine compete for vitamin B_{12} and damage the intestinal mucosa.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

84) Which of the following may accentuate variations leading to appearance of new species and hence evolution ?

(1) No mutation in population

(2) No gene recombination in population

(3) Random mating in population

(4) Habitat fragmentation

85) **Assertion (A)** : A foreign DNA and plasmid cut by same restriction endonuclease can be joined to form a recombinant plasmid using Taq polymerase
Reason (R) : Selectable marker is a gene, whose expression helps to identify transformed cell.

(1) Assertion (A) and Reason (R) both are correct but Reason (R) is the correct explanation of Assertion (A)

(2) Assertion (A) and Reason (R) both are correct but Reason (R) is not the correct explanation of Assertion (A)

(3) Assertion (A) is correct but Reason (R) is not correct

(4) Assertion (A) is not correct but Reason (R) is correct

86) Match column-I with column-II, and choose the correct combination from the option given below :

	Column-I		**Column-II**
(a)	Molecular scissor	(i)	Large scale production of products
(b)	Plasmid	(ii)	Recombinant protein
(c)	Bioreactors	(iii)	pBR 322
(d)	Heterologous host	(iv)	Eco RI

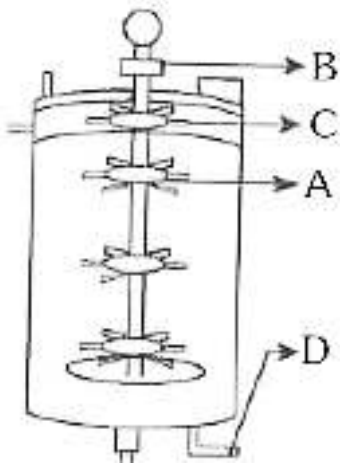
(1) a-iii, b-iv, c-ii, d-i

(2) a-iv, b-iii, c-i, d-ii

(3) a-i, b-ii, c-iii, d-iv

(4) a-ii, b-i, c-iv, d-iii

87) Identify the part of a bio-reactor which is used as a foam braker from the given figure.



(1) A

(2) B

(3) D

(4) C

88) Match the column-I and column-II :

	Column-I		**Column-II**

(A)	Alpha-1 antitrypsin	(i)	Treatment of emphysema
(B)	Transgenic mice	(ii)	Produce human protein-enriched milk
(C)	Rosie	(iii)	Model for cancer research
(D)	Toxicity testing	(iv)	Study of drug safety using animal

	A	**B**	**C**	**D**
(1)	(iv)	(iii)	(ii)	(i)
(2)	(i)	(ii)	(iii)	(iv)
(3)	(ii)	(iii)	(i)	(iv)
(4)	(i)	(iii)	(ii)	(iv)

(1) 1

(2) 2

(3) 3

(4) 4

89) Which of the following statement is incorrect ?

(1) *Bacillus thuringiensis* forms cry protein crystals during a particular phase of their growth.

(2) The proteins encoded by the genes cryIAc and cryIIAb control corn borer.

(3) RNAi takes place in all eukaryotic organisms as a method of cellular defense.

(4) The choice of cry genes depends upon the crop and the targeted pest.

90) Select incorrect statement :-

(1) Biofertilizers are organisms that enrich the nutrient quality of the soil.

(2) Rhizobium and Azotobacter form symbiotic association with roots of leguminous plants.

(3) Many members of the genus *Glomus* form mycorrhiza.

(4) In paddy fields, Cynobacteria serve as an important biofertilizer.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	3	4	3	3	4	1	1	2	2	3	2	4	1	3	1	3	2	3	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	1	2	1	4	2	2	3	3	4	2	2	1	4	2	1	1	1	4	1
Q.	41	42	43	44	45															
A.	4	1	4	1	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	2	2	3	2	2	4	3	3	1	1	4	4	4	2	3	1	1	2	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	3	1	3	1	2	1	4	4	1	1	4	4	2	3	4	2	1	1
Q.	86	87	88	89	90															
A.	4	3	3	4	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	4	3	3	4	1	1	4	2	1	3	4	1	4	3	3	3	3	3	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	2	3	3	1	1	3	3	4	3	3	1	3	1	3	2	3	4	1	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	3	4	4	2	3	4	1	1	1	3	1	3	4	1	3	2	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	1	4	2	1	2	3	3	4	3	3	4	1	3	1	2	2	2	2	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	3	1	4	4	2	4	4	2	2										

SOLUTIONS

PHYSICS

1) $LC = \frac{0.5mm}{100} = 0.005$ mm

Let N^{th} CSD coincides with reference line

Main scale reading = 12 divisions = 12 × 0.5 mm = 6mm

Corrected reading = MSR + N × LC – ZE

6.255 mm = 6 mm + N (0.005 mm) – 0.010 mm

N (0.005mm) = 0.265 mm

$\frac{0.265}{0.005} = 53^{rd}$division

2) Let W be the water equivalent of calorimeter.

Heat lost by metal = Heat gained by water + Calorimeter

200 × 0.1 × (100 – 30) = (250 + W) × (30 – 25)

⇒ (250 + W) = 280 ⇒ W = 30 g

3) $Y = \frac{4F}{\pi D^2}\frac{L}{x}$

$= \frac{(4)(50)(1.57)}{(\pi)\left(10^{-6}\right)\left(5\times10^{-4}\right)} = 20\times10^{10}N/m^2$

Also $\frac{\Delta Y}{Y} = \pm\left(\frac{2D}{D}+\frac{\Delta x}{x}+\frac{\Delta L}{L}\right)$

$\frac{\Delta Y}{Y} = \pm\left(2\times\frac{0.001}{0.100}+\frac{0.001}{0.050}+\frac{0.1}{157.0}\right)$

$\frac{\Delta Y}{Y}$ = ± (0.02 + 0.02 + 0.0006)

$\Delta Y = \pm$ (0.0406) × 20 × 10^{10} = ± 8.1 × 10^9 N/m^2

4) By half deflection method, resistance of galvanometer $G = \frac{RS}{R-S}$

$G_1 = \frac{96\times3000}{3000-96} = \frac{288000}{2904} \simeq 99\Omega$

$G_2 = \frac{98\times4900}{4900-98} = \frac{480200}{4802} = 100\Omega$

From above $G_{avg} = \frac{99+100}{2} = 99.5\Omega \approx 100\Omega$

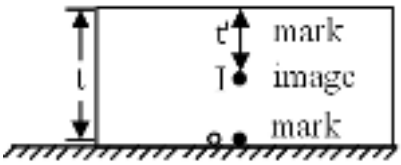
5) δ_{min} = 2i – A ⇒ A = 2i –δ_{min}

2 × 53 – 46 = 60°

$\mu = \frac{\sin\left(\frac{A+\delta_{min}}{2}\right)}{\sin\frac{A}{2}} = \frac{\sin\left(\frac{60+46}{2}\right)}{\sin\left(\frac{60}{2}\right)}$

$$\mu = \frac{\sin 53^\circ}{\sin 30^\circ} = \frac{8}{5} = 1.6$$

$$v = \frac{c}{\mu} = \frac{3 \times 10^8}{1.6} = \frac{15}{8} \times 10^7 \text{m/s} = 1.9 \times 10^7 \text{m/s}$$

6) 

$$LC = \left(\frac{50-49}{50}\right) \times \frac{1}{20}\text{cm} = 0.001\text{cm}$$

$R_A = 8.25 + 15\ (0.001)\ \text{cm} = 8.265\ \text{cm}$

$R_B = 6.57 + 35\ (0.001)\ \text{cm} = 6.605\ \text{cm}$

$R_C = 3.85 + 5\ (0.001)\ \text{cm} = 3.855\ \text{cm}$

$t = R_A - R_C$, $t' = R_B - R_C$

$$\Rightarrow \mu = \frac{t}{t'} = \frac{R_A - R_C}{R_B - R_C} = \frac{8.265 - 3.855}{6.605 - 3.855} = 1.60$$

7) Rise in temperature generates more electron - hole pairs, thus conductivity increases and for same voltage across P-N junction diode, current increases, effectively lowering the dynamic resistance.

8)

For capacitor C after inserting dielectric of dielectric constant 2

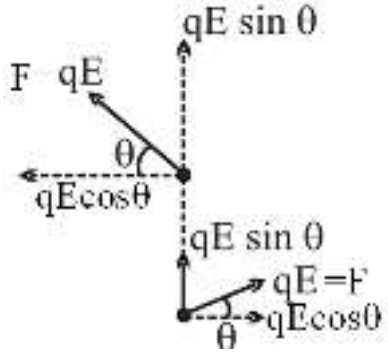
$C' = 2C$ and $V' = \frac{V_0}{2}$

and for 2C after inserting dielectric of dielectric constant 3

$C'' = 6C$ and $V'' = \frac{V_0}{3}$

$$V_{common} = \frac{C'V' + C''V''}{C' + C''}$$

$$= \frac{(2C)\left(\frac{V_0}{2}\right) + (6C)\left(\frac{V_0}{3}\right)}{2C + 6C} = \frac{3}{8}V_0$$

9) 

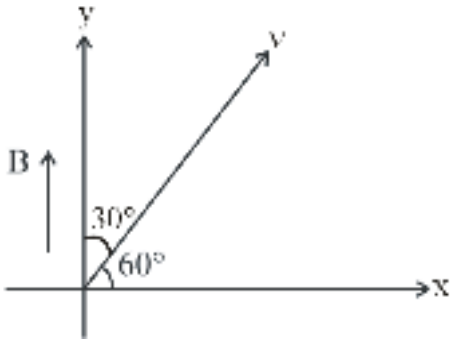
10) $U = \frac{k}{a}(q_1q_2 + q_1q_4 + q_2q_3 + q_3q_4)$.

$$+ \frac{k}{a\sqrt{2}}(q_1 q_3 + q_2 q_4)$$

$$= \frac{k}{a}[(q)(-q) + (q)(-q) + (-q)(q) + (q)(-q)]$$

$$+ \frac{k}{a\sqrt{2}}[(+q)(+q) + (-q)(-q)]$$

$$= \frac{kq^2}{a}\left[\frac{2}{\sqrt{2}} - 4\right] = -\frac{2.6q^2}{4\pi\varepsilon_0 a}$$

11) 

Path of proton will be helix

$$r = \frac{m\nu \sin 30^\circ}{qB}$$

$$r = \frac{1.67 \times 10^{-27} \times 2 \times 10^6 \times \frac{1}{2}}{1.6 \times 10^{-19} \times 0.104}$$

r = 0.1 m

$$T = \frac{2\pi m}{qB} = \frac{2\pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.104}$$

$T = 2\pi \times 10^{-7}$ sec.

12)

$B = \mu H$

$= (8 \times 10^{-3})$ (160) Wb/m^2

$= 1.28\ Wb/m^2$

13)

This question is based on drift velocity $\left[V_d = \frac{i}{neA}\right]$

$$V_d = \frac{20}{10^{29} \times 1.6 \times 10^{-19} \times 10^{-6}}$$

$V_d = 1.25 \times 10^{-3} m/s$

14) $e_{ind} = M\frac{dI}{dt}$

$e_{ind} = (M)1000\pi \cos(100\pi t)$

$e_{ind} = 5\pi = (1000\pi)\ M$ (compare max. values)

$M = 5 \times 10^{-3}$ H

M = 5 mH

15) Here, L = 2H, C = 32μF = 32×10^{-6} F, R = 10Ω

 Resonance frequency,

$$\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2 \times 32 \times 10^{-6}}} = \frac{10^3}{8}\,\text{rad}\,\text{s}^{-1}$$

Quality factor = $\frac{\omega_r L}{R} = \frac{10^3 \times 2}{8 \times 10} = 25$

16)

Infrared waves of EM spectrum has heating capability and these wave heat earth surface.

17)

L = 25 cm = $v_0 + f_e$

$25 = v_0 + 2.5$

$v_0 = 22.5$ cm

$f_0 = 1.5$ cm

$$\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0}$$

$$\frac{1}{22.5} - \frac{1}{u_0} = \frac{1}{1.5}$$

$$\frac{1}{u_0} = \frac{1}{22.5} - \frac{1}{1.5} = \frac{10 - 150}{225}$$

$$u_0 = \frac{-225}{140}$$

$$\text{M.P.} = \frac{v_0}{u_0} \times \frac{D}{f_e} = \frac{22.5 \times 140 \times 25}{225 \times 2.5} = 140$$

18)

Path difference for dark-fringes in YDSE, $\Delta x = (2n - 1)\frac{\lambda}{2}$

Path difference $\Rightarrow \Delta x = (2 \times 3 - 1)\frac{\lambda}{2} = \frac{5\lambda}{2} = \frac{5 \times 0.6\mu m}{2} = 1.5\ \mu m$

19) After first, $I_1 = \frac{I_0}{2}$

After second, $I_2 = I_1 \cos^2 30°$ (By Malus law)

$= \frac{I_0}{2} \times \frac{3}{4} = \frac{3I_0}{8}$

After third, $I_3 = I_2 \cos^2 30°$

$= \frac{3I_0}{8} \times \frac{3}{4} = \frac{9I_0}{32}$

After fourth, $I_3 \cos^2 30°$

$= \frac{9I_0}{32} \times \frac{3}{4} = \frac{27}{128} I_0 = 0.21\ I_0$

20) $\frac{mv^2}{r} = \frac{e^2}{4\pi \in_0 r^2} \Rightarrow r = \frac{e^2}{4\pi \in_0 mv^2}$

21) From quantization condition

$mvr = \frac{nh}{2\pi}$ for n = 1

or $mv = \frac{h}{\lambda} = \frac{h}{2\pi r}$

or $\lambda_{de\text{-}Broglie} = 2\pi r = 1.058\ \pi$ Å

22)

By Einstein's Equation

$E = \phi_0 + K_{max}$

$\Rightarrow K_{max} = E - \phi$

$= hv - \phi$

$= (6.6 \times 10^{-34})(8 \times 10^{14}) - 3.2 \times 10^{-19}$

$= 2.1 \times 10^{-19}$ Joule

23)

- Internal Energy Change:

$\Delta U = nC_V \Delta T, \quad C_V = \frac{3}{2}R$

- Work Done:

$W = \int PdV$

- First Law of Thermodynamics:

$Q = \Delta U + W$

For Helium, degrees of freedom, f = 3

Change in internal energy, $U = \frac{f}{2}\Delta(PV)$

$\Longrightarrow \Delta U = \frac{3}{2} \times (4p_o\ V_o - p_o\ V_o) = \frac{9}{2} p_o\ V_o$

Work done by the gas is equal to the area under $P - V$ curve. Here, it is a trapezium under the curve.

$W = \frac{1}{2}\ V_o\ (p_o + 2p_o) = \frac{3}{2} p_o\ V_o$

From first law of thermodynamics, total heat given is:

$$\Delta Q = W + \Delta U$$

$$\Delta Q = \frac{3}{2} p_0 V_0 + \frac{9}{2} p_0 V_0$$

$$\Delta Q = 6 p_0 V_0$$

24) $$\frac{n_1 + n_2}{\gamma_{mix} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}$$

25)

The area of I - λ graph gives the total intensity of radiation which will be proportional to the 4th power of temperature of body in kelvin.

Area $\propto T^4$

$$\frac{A_1}{A_2} = \frac{(327 + 273)^4}{(627 + 273)^4}$$

$$\frac{A_1}{A_2} = \left(\frac{600}{900}\right)^4$$

Then

$$\Rightarrow \frac{A_1}{A_2} = \left(\frac{2}{3}\right)^4 \Rightarrow \frac{A_1}{A_2} = \frac{16}{81}$$

26)

For parallel combination of two rods of equal length and equal area of cross-section :

$$K = \frac{K_1 + K_2}{2} = \frac{K_1 + \frac{4K_1}{3}}{2} = \frac{7K_1}{6}$$

Hence, $$\frac{K}{K_1} = \frac{7}{6}$$

27) $\gamma_{app} = \gamma_{real} - \gamma_{vessel}$

Given that

$\gamma_{real} = 7 \cdot \gamma_{vessel}$

$\gamma_{app} = \gamma_{real} - \gamma_{vessel}$

$\gamma_{app} = 7\gamma_{vessel} - \gamma_{vessel} = 6\gamma_{vessel}$

$$\text{Ratio} = \frac{\gamma_{real}}{\gamma_{app}} = \frac{7\gamma_{vessel}}{6\gamma_{vessel}} = \frac{7}{6}$$

28)

$U = U_0(1-\cos\alpha x)$

$$F = -\frac{dU}{dx}$$

$ma = -U_0\alpha \sin\alpha x$

$$a = -\left(\frac{U_0\alpha}{m}\right)\sin\alpha x$$

$\sin\alpha x \simeq \alpha x$

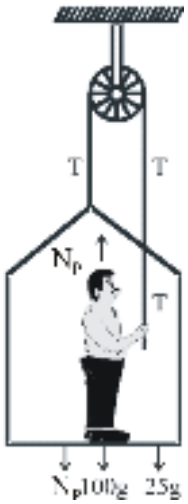
$$a = -\frac{U_0\alpha^2}{m}x \Rightarrow \omega = \sqrt{\frac{U_0\alpha^2}{m}}$$
$$T = \frac{2\pi}{\alpha}\sqrt{\frac{m}{U_0}}$$

29)

$m = K E^a v^b F^c$
$[M^1L^0T^0] = [M^1L^2T^{-2}]^a [L^1T^{-1}]^b [M^1L^1T^{-2}]^c$
$\Rightarrow a + c = 1$
$2a + b + c = 0$
$-2a - b - 2c = 0$
On solving three equation we get
$a = 1$; $b = -2$; $c = 0$
So, $m \propto Ev^{-2}$

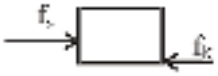
30)

distance travelled in t = 0 to t = 10 sec
$= \frac{1}{2} \times 10 \times 10 = 50$
distance travelled t = 10 to t = 20 sec
$= \frac{1}{2} \times 10 \times 10 = 50$
Avg. speed is same in t = 0 to t = 10.
& t = 10 to t = 20 sec



31)
equation for painter & crate system.
$2T - 125\,g = 125\,a$ .... (1)
equation for painter
$N_p + T - 100\,g = 100\,a$ ($N_p = 450$ N)
$T - 550 = 100\,a$ ..... (2)
from eqation (1) & (2)
$a = 2$ m/sec^2

32) $f_s = \mu_s mg = 0.5 \times 40 \times 10 = 200$ N



$f_K = 0.4 \times 40 \times 10 = 160$ N

$f_s - f_k = 40a$

$200 - 160 = 40a$

33)

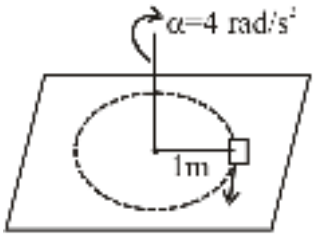
From work-energy theorem

$$W_F + W_{mg} + W_N = \frac{1}{2}mv^2$$

$$F \times R + mgR + 0 = \frac{1}{2}mv^2$$

$$5 \times 5 + \frac{1}{2} \times 10 \times 5 = \frac{1}{2} \times \frac{1}{2} \times v^2 \quad \Rightarrow \quad v = \sqrt{200} = 14.14 \text{ m/s}$$

34)



m = 2kg

a_c = ?

$\omega_0 = 0$, r = 1m

$\alpha = 4$ rad/s^2

t = 1 sec.

$\omega = \omega_0 + \alpha t$

$\omega = 0 + 4(1) = 4$ rad/s

$a_c = \omega^2 r = (4)^2 \times (1) = 16$ m/s^2

35) Velocity on hitting the surface

$= \sqrt{2 \times 10 \times 5} = 10$ m/s

Velocity after first bounce $v = \frac{3}{4} \times 10$

Time taken from first bounce to the second bounce

$$= \frac{2v}{g} = 2 \times \frac{3}{4} \times 10 \times \frac{1}{10} = 1.5 \text{ s}$$

36) $\tau = I\alpha$

$\Rightarrow \vec{r} \times \vec{F} = I\alpha$

$\Rightarrow rF \sin 90° = I\alpha$

$\Rightarrow rF = (mr^2)\alpha$

$$\Rightarrow \alpha = \frac{F}{mr} = \frac{30}{3 \times 0.4}$$

$= 25$ rad/s^2

37)

$L_f = 0 \Rightarrow L_i = 0$

$mv^0R - \frac{mR^2}{2}\omega^0 = 0$

$\Rightarrow \frac{v_0}{r\omega_0} = \frac{1}{2}$

38)

At maximum height of water rise in tank, volume flow rate into tank = Volume flow rate out from tank.

$Q = A.v$

$70 = 1 \times \sqrt{2gh}$

$\Rightarrow h = \frac{70 \times 70}{2 \times 980} = 2.5$ cm

39) $V_T = \frac{2r^2(\rho_b - \rho_L)g}{\eta}$

$V_{T_1} = V_{T_2}$

$r_1^2\left(6 \times 10^3\right) = r_2^2\left(9 \times 10^3\right)$

$\frac{r_1}{r_2} = \sqrt{\frac{3}{2}}$

40) $-\frac{3}{2}\frac{GMm}{R} + \frac{1}{2}mv^2 = -\frac{GMm}{R}$

$\Rightarrow v = \sqrt{\frac{GM}{R}}$

41) $v_{rel} = v_A - v_B = 10 - 5 = 5$ m/s

For 1st meeting $S_{rel} = 5$m

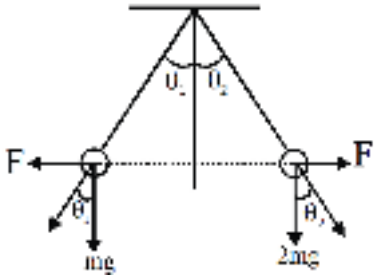
time taken, $t_1 = \frac{S_{rel}}{v_{rel}} = \frac{5}{5}$

$t_1 = 1$ sec.

For 2nd meeting, $S_{rel} = 15$ m

$t_2 = \frac{S_{rel}}{v_{rel}} = \frac{15}{5} = 3$ sec

Total time = $t_1 + t_2 = 1 + 3 = 4$ sec

42) 

It is cleare from figure that θ_1 is greater than θ_2

43)

$6 = 10E - 12(10\ r)$ ...(1)

$0 = 10E - 14(10\ r)$ ...(2)

So, $6 = 10E - 12\left[10 \times \frac{E}{14}\right]$

$E = \frac{42}{10} = 4.2V$

44) Conversion of proton into neutron is endothermic. Inside the nucleus this is possible but not in free state as the requirement of energy.

45)

All the particles having phase difference 2π or integral multiples of 2π will have same phase.

CHEMISTRY

46)

A. For first-order reactions:
B. Half-life: $t_{1/2} = 0.693/k$, independent of initial concentration → Graph 1 is correct.
C. Integrated rate law: $\ln[A]_t = -kt + \ln[A]_0$, gives a straight line with slope –k →Graph 4 shows slope = +K, which is incorrect; slope should be –k.
D. Plot of $\log(A_0/A_t)$ vs time gives slope k/2.303 → Graph 2 is correct.
E. Concentration vs time curve is exponential decay → Graph 3 is correct.

47)

A → excess

$r = \frac{dx}{dt} = k[B]^n$

$\frac{r}{2r} = \left[\frac{0.1}{0.4}\right]^n \Rightarrow n = \frac{1}{2}$

48) $r = 0.529\ \frac{n^2}{Z}$ Å

$$\frac{0.529}{Z}\left[(n+1)^2 - n^2\right] = \frac{0.529}{Z} \times 1^2 \times 5$$

$(n + 1)^2 - n^2 = 5$

$\Rightarrow n = 2$

49)

$\lambda_{eq}\ BaSO_4 = \lambda_{eq}\ BaCl_2 + \lambda^{\infty}_{eq}\ H_2SO_4 - \lambda_{eq}\ HCl$

$\lambda_{eq}\ BaSO_4 = x_1 + x_2 - x_3$

$\lambda_m = \lambda_{eq} \times V.F. = 2\ (x_1 + x_2 - x_3)$

$$\lambda_m = \frac{K \times 10^3}{S}$$

$$S = \frac{Y \times 10^3}{2(x_1 + x_2 - x_3)}$$

$$KSP\ (BaSO_4) = S^2 = \frac{Y^2 10^6}{4(x_1 + x_2 - x_3)^2}$$

50)

Standard reduction potential of Au is greater than Cu^{+2}, Fe^{+2}, Ni^{+2}, Cr^{+3} and Zn^{+2} (given metals) therefore they will replace Au from its salt solution.

51) $A + D \rightleftharpoons AD$, K_1 ..... (1)

$AD + D \rightleftharpoons AD_2$, K_2 ..... (2)

$AD_2 + D \rightleftharpoons AD_3$, K_3 ..... (3)

$A + 3D \rightleftharpoons AD_3$, K .....(4)

(4) = (1) + (2) + (3)

$K = K_1 \times K_2 \times K_3$

$\log K = \log K_1 + \log K_2 + \log K_3$

52) Option (1)

m. eq. of $H_2SO_4 = N \times (V)\ mL = M \times V.f \times V\ (mL)$

$$\frac{1}{10} \times 2 \times 200 = 40$$

m. eq. of NaOH $= \frac{1}{10} \times 800 = 80$

$\Rightarrow$ m. eq. of NaOH > m. eq. of H_2SO_4

Solution is basic $\Rightarrow [OH^-] = \frac{80 - 40}{1000}$

Option (2)

m. eq. of HCL $= \frac{1}{10} \times 800 = 80$

m. eq. of $Ba(OH)_2$ $\frac{1}{10} \times 2 \times 200 = 40$

$\Rightarrow$ m. eq. of of HCl > m. eq. of $Ba(OH)_2$

$\Rightarrow$ Solution is acidic

Option (3)

m. eq. H_2SO_4 = m. eq. of $Ba(OH)_2$
⇒ Solution is neutral
Option (4)

m. eq. of $HNO_3 = \frac{1}{10} \times 250 = 25$

m. eq. of KOH $= \frac{1}{10} \times 750 = 75$

$[OH^-] = \frac{50}{1000}$ (maximum)

⇒ $[OH^-]\uparrow$, $P^{OH}\downarrow$, $P^{H}\uparrow$

53) **1.** For A) $AlCl_3$ with α = 0.8 ; (dissociation)
$AlCl_3$ dissociates into 4 ion (Al^{3+} and $3Cl^-$. so n = 4
i = 1 + (4– 1)(0.8) = 1 + 3(0.8) = 1 + 2.4 = 3.4
This matches (P)
2. For B) $BaCl_2$ with α = 0.9 ; (dissociation)
$BaCl_2$ dissociates into 3 ions (Ba^{2+} and $2Cl^-$). So, n = 3
i = 1 + (3 – 1)(0.9) = 1 + 2(0.9) = 1 + 1.8 = 2.8
This matches (Q)
3. for C) CH_3COOH in benzene (Association)
Acetic acid in benzene forms a dimer, so 2 molecules associate (m = 2). The van't Hoff factor should be less than 1,
so i = 0.5i = 1 + (4 – 1)(0.9) = 1 + 3(0.9) = 1 + 1.7 = 3.7
This matches (S)
4. for D) $K_4[Fe(CN)_6]$ with α = 0.7 ; (dissociation)
The complex dissociates into 5 ions : ($4K^+$ and $1[Fe(CN)_6]^{4-}$). so, n = 5
i = 1 + (5 – 1)(0.7) = 1 + 4(0.7) = 1 + 2.8 = 3.8
This matches (R)

54)

Boiling point of solution
$= 100 + \Delta T_b$
Freezing point of solution = $0 - \Delta T_f$
Difference in temperature (given)
$= 100 + \Delta T_b - (-\Delta T_f)$
$103.57 = 100 + \Delta T_b + \Delta T_f$
$= 100 + \text{molality} \times (K_b + K_f)$
$= 100 + \text{molality} \times (0.52 + 1.86)$

$\therefore \text{molality} = \frac{103.57 - 100}{2.38} = \frac{3.57}{2.38} = 1.5m$

and molality $= \frac{\text{moles} \times 1000}{W_{gm(solvent)}}$;

$1.5 = \frac{\text{moles} \times 1000}{500}$ = 0.75 moles

55)

$C_xH_y + (x + y/4)\ O_2 \rightarrow x\ CO_2 + (y/2)\ H_2O$
Moles of CO_2 = x × moles of hydrocarbon
Moles of H_2O = (y/2) × moles of hydrocarbon
 Moles of hydrocarbon = 0.1
Moles of CO_2 = 0.1 → x × 0.1 = 0.1 → x = 1
Moles of H_2O = 0.2 → (y/2) × 0.1 = 0.2 → y/2 = 2 → y = 4
Hydrocarbon formula = CH_4

56)

K_2MnO_4 2 + x – 8 = 0 , x = +6
CaO_2, 2 + 2x = 0 , x = –1
$H_2S_2O_7$ 2 + 2x – 14 = 0 , x = +6
Fe_3O_4 –2 , $Cr_2O_7^{2-}$, x = +6
NO_3^- x – 6 –1 , x = +5

57)

$C_6H_{6(\square)} + \frac{15}{2} O_{2(g)} \rightarrow 6\ Co_{2(g)} + 3H_2O_{(\square)}$
ΔH = –45.9 – 6 × 393.5 – 3 × 285.9
= –3264.6 KJ

58)

$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$
2x x
$K_p = (2x)^2x = 4x^3$
$\Delta G° = -RT \ln K_p$
$\Delta G° = -RT \ln 4x^3$
$= -RT \ln 4 - 3RT \ln x$

59)

Work done by the gas: W = – 8.31KJ = – 8310 Joule
$W = -2.303\ nRT \log \frac{V_2}{V_1}$
$\Rightarrow +8310 = +2.303 \times 1 \times 8.31 \times T \times \log \frac{30}{3}$
$8310 = 8.31 \times T \times 2.303 \times (\log_{10}10)$
$T = 8310 / (8.31 \times 2.303)$
$\Rightarrow T = \frac{1000}{2.303} = 434.2K$

60)

$K_p = K_c\ (RT)^{\Delta ng}$

If $\Delta ng < 0 \rightarrow K_p < K_c$

1. $\mathbf{N_2O_3 \rightleftharpoons NO_2 + NO}$

$\Delta ng = (1 + 1) - 1 = +1 \rightarrow K_p > K_c$

2. $\mathbf{2H_2 + O_2 \rightleftharpoons 2H_2O}$

$\Delta ng = 2 - (2 + 1) = -1 \rightarrow K_p < K_c$

3. $\mathbf{H_2 + I_2 \rightleftharpoons 2HI}$

$\Delta ng = 2 - 2 = 0 \rightarrow K_p = K_c$

4. $\mathbf{2SO_3 \rightleftharpoons 2SO_2 + O_2}$

$\Delta ng = (2 + 1) - 2 = +1 \rightarrow K_p > K_c$

61)

Fact

62)

(A) In < Tl → IE ↓, EN ↓, Size ↑, Same group → P, Q, S, T
(B) F > Cl → IE ↑, EN ↑, same group → P, Q, T
(C) Na^+ > Ne → Na^+, Ne are isoelectronic (10 electrons); IE ↑, EN ↑ → P, Q, R
(D) Al > Ga → Size ↑ due to d-block contraction; same group → S, T

63) NCERT XI, Ch-4, Part-I

64) N–N single bond is much weaker than P–P due to lone pair repulsion and small size of N atoms. Hence, statement (2) is incorrect.
Final Answer: 2

65) NCERT Pg. No. # 116, 117, Part-I

66) Both acetone & acetonitrile are polar substance.

67) NCERT XI, Ch-4, Part-I

68) x + o + (–2) + (–2) = –1
(i) x = +3
(ii) CN = 6
(iii) [+3 – 6] ⇒ –3
Cr Cl

69) $[Co(en)_2Cl_2]^{\oplus}$ → 2Gl[cis + trans], 2Ol[enantiomers]

70) NCERT Pg. No. # 131

Greater is the electron pair donating ability high is the stability of complex.

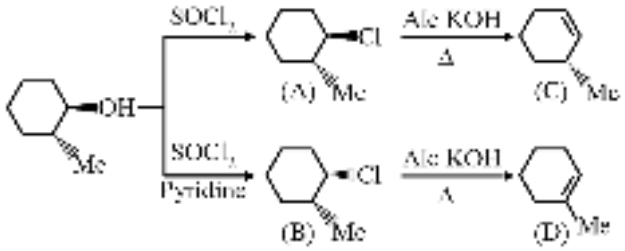
71) NCERT Pg. No. # 209

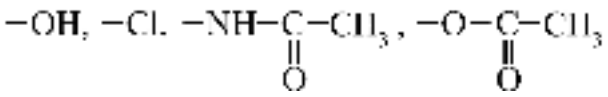
72) $\alpha < \beta$ - Bond angle
Al is not $e^{\ominus}$ deficient.

73) $Pb_3O_4 \rightarrow 2\overset{+2}{P}bO + \overset{+4}{P}bO_2$, $Fe_3O_4 \rightarrow FeO + Fe_2O_3$
$Mn_3O_4 \rightarrow \overset{+2}{M}nO + \overset{+3}{M}n_2O_3$, $Co_3O_4 \rightarrow \overset{+2}{C}oO + \overset{+3}{C}o_2O_3$

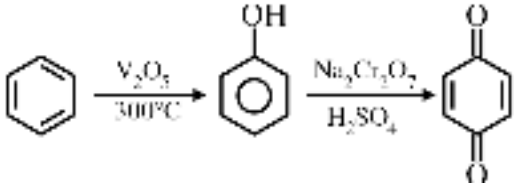
74) $Ce^{+4} \xrightarrow{+e^-} Ce^{+3}$
↓
good oxidising agent

75) NO_2^- & Br^- gives respectively NO_2 & Br_2 having same colour.

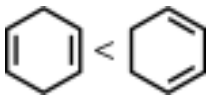
76) 

77) 

are o/p directing

78) 

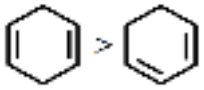
79)

(i)  Stability of alkene

Reso

(ii)  Heat of combustion

'5C' '4C'

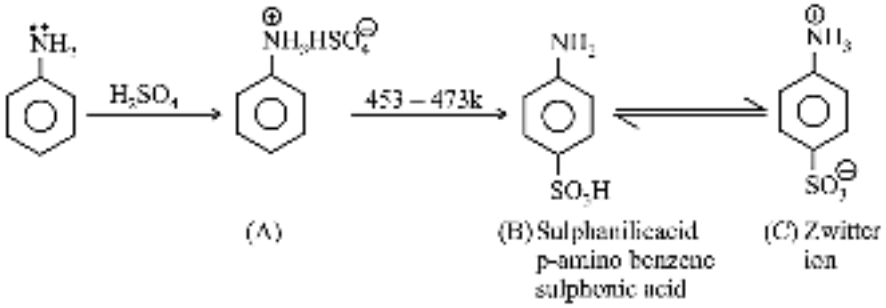
(iii)  Heat of hydrogenation

(Reso)

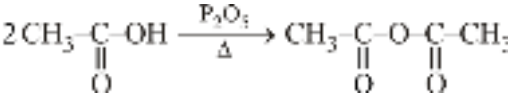
(iv) $CH_2=CH_2 < CH_2=CH-CH=CH_2$ Bond length

(Reso)

80)



Final answer: option (2)

81) 

82)

Option	Reaction Equation	Validity	Correct Product / Reason
1	$CH_3CH_2OH \xrightarrow{alk.KMnO_4} CH_3COOH$	Correct	Primary alcohols are oxidized to acids by strong oxidizing agents.
2	$CH_3CONH_2 \xrightarrow{H_3O^+} CH_3COOH$	Correct	Amides undergo acidic hydrolysis to yield carboxylic acids.
3	$CH_3CN \xrightarrow{Complete\ hydrolysis} CH_3COOH$	Correct	Complete hydrolysis of nitriles yields the corresponding carboxylic acid
4	$CH_3COCl \xrightarrow{H_2O} CH_3CHO$	Incorrect	Hydrolysis of acid chlorides yields CH_3 COOH+HCl, not an aldehyde.

Final answer : 4

83)

NCERT-XII, Pg. # 414, Part-2

84) NCERT pg No.271

85) NCERT Pg No. 268

86)

NCERT-XI, Pg. # 368, Part-2

87)

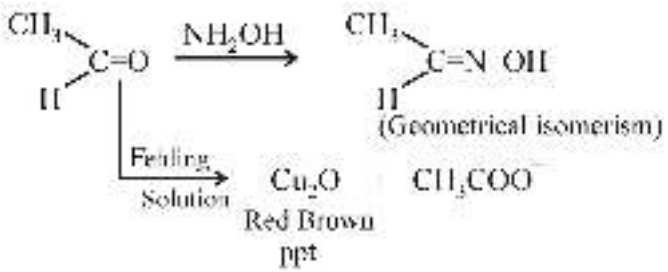
Carboxylic anhydride → $C_nH_{2n-2}O_3$

88)

I, II, III have same molecular formula

89)

It's a fact



90)
Final answer : 2

BIOLOGY

91) NCERT XI, Pg. # 8

92)

NCERT-XI, Pg # 23, 24

93) NCERT XI Pg. # 26,27
(2.6)

94)

NCERT XI Pg. # 14,15

95) NCERT Pg.#26-33.

96)

NCERT Page No. 64

97) NCERT Pg. No. # 72

98) NCERT Pg.# 76.

99)

NCERT-XI, Page No. 94 Para 6.8

100)

NCERT XI Pg. # 93,94,99
Solution : (a) True – Prokaryotic flagella are simple, made of flagellin, show rotary movement, while eukaryotic flagella are complex (9+2 microtubule arrangement) and show bending movement.
(b) False – Middle lamella is mainly made of calcium pectate, not calcium carbonate.
(c) True – Fluid mosaic model explains that lipids in the bilayer are quasi-fluid, allowing lateral movement of proteins.
(d) False – Polar molecules cannot easily cross the non-polar lipid bilayer; they need carrier proteins or channels.
Final answer : Two

101)

Explain the Question : The question asks to identify the correct labels for A, B, and C in the given cellular secretion pathway flow chart starting from RER (Rough Endoplasmic Reticulum) to extracellular fluid through the Golgi apparatus and vesicles.
Concept : This question is based on Intra - Endomembrane System (ER, GB, Lysosome, Vacuole)
Solution :
A = ER vesicle (carries proteins from RER to Golgi).
B = Secretory vesicle (carries processed proteins from Golgi to membrane).
C = Plasma membrane (where secretion occurs.
Final answer : 3

102)

NCERT XI, Page # 123, 124

103)

Explanation:- The question has provided an image depicting a sequence of stages during meiosis and asks to choose the correct statement describing one of the labeled stages (A, B, C, D).

Concept:- Meiosis I

Solution:- A-Crossing overThis is because the diagram illustrates the process of meiosis, specifically the stages where crossing over occurs. Crossing over is the exchange of genetic

material between homologous chromosomes, leading to genetic variation.

Final answer:- A-Crossing over

104)

Explain Question –
The question asks you to identify which one of the four given statements is not true about cell division/cytokinesis and the cell cycle.

Concept – Cell Cycle and Cell Division

Solution –

A. True. The cell plate grows outward and later forms the middle lamella between daughter cell walls.

B. True. organelles like mitochondria and plastids divide and are distributed to daughter cells during cytokinesis.

C. True. In plants cytokinesis is centrifugal (cell plate formation). In animals it is centripetal (furrow/cleavage).

D. Not true. Interphase lasts about 95% of the cell cycle, not M-phase. M-phase is very short.

Final answer – Option 4

105) NCERT XI Pg. # 114, 115

106)

Explain Question – The question asks to identify the correct Biomolecule - function/nature pair from the given options.
Concept – Function/Nature Of Biomolecule
Solution –
Collagen → structural protein, not for fighting infections → incorrect
GLUT-4 → glucose transporter, not a hormone → incorrect
Inulin → storage carbohydrate, polymer of fructose → correct
Trypsin → digestive enzyme, not a carbohydrate → incorrect
Final Answer – Option 3

107) NCERT XI Pg. # 108

108) NCERT XI Pg. # 173, 174

109) NCERT XI Pg # 175 -178

110)

Explaining the Question:
This question asks to select the correct statement related to factors affecting photosynthesis.

Concept:
Factors affecting photosynthesis.

Solution:
Tomatoes and bell peppers show increased photosynthetic rate and higher yield when grown in a CO_2-enriched atmosphere because CO_2 is a major limiting factor for photosynthesis in C_3 plants under normal conditions.

Final Answer: Option 2

111)

Explain Question – The question is asking you to evaluate two statements related to photosystems and cyclic photophosphorylation, and identify which option correctly reflects their validity.

Concept – Based on cyclic photophosphorylation and structure of photosystems.

Solution – Statement I: Correct. When only PSI is functional, electrons cycle back within the photosystem, leading to ATP formation through cyclic photophosphorylation.

Statement II: Correct. Each photosystem contains a light-harvesting complex (antennae) made of pigments, except for one reaction center chlorophyll-a molecule.

Final answer – Option 4

112)

NCERT (XI) Pg. # 154, 157, 160, 164, 165

113) NCERT XI, Pg. # 158

114) NCERT Pg. # 9, 10, 11

115) NCERT-XII, Pg. # 18

116)

NCERT XII Pg.# 29

117) NCERT XII Pg.# 58

118)

NCERT-XII, Pg. # 67

119) NCERT-XII, Pg. # 61

120)

NCERT-XII, Pg. # 67
Explain Question: The question asks to find the given genes.
Concept: Gene Mapping (Linkage)
Solution: To determine the gene sequence, we check whether the given map distances fit on a linear chromosome.
Given distances

A. W–Y = 18
B. W–X = 26
C. W–Z = 40
D. X–Y = 8
E. X–Z = 14

Step 1: Place X, Y, and W
Since:

A. X–Y = 8
B. W–Y = 18
C. W–X = 26

Check if:
X — Y — W fits the distances.
Distance X→Y = 8
Distance Y→W = 18
So X→W = 8 + 18 = 26, which matches the given value.
Thus, the order so far is:
X — Y — W
Step 2: Place Z
We try to place Z relative to X and W.
Given:

A. X–Z = 14
B. W–Z = 40

Try placing Z on the X side:
If the order is:
Z — X — Y — W
Distances would be:

A. X–Z = 14
B. X–Y = 8

C. Y–W = 18

Thus W–Z =(X–Z) + (X–W) = 14 + 26 = 40, which matches the given value.
So, the correct answer is: Z — X — Y — W
Final answer: Option 3

121)

NCERT XII, Pg. # 70

122)

NCERT-XII, Pg. # 54

123) NCERT XII Pg.# 94

124) NCERT XII, Pg. # 94

125)

NCERT XII, Pg. # 98

126)

NCERT XII, Pg. # 96

127) NCERT XII Pg. # 84

128) NCERT XII, Pg. # 105

129) NCERT XII Pg.# 86

130) NCERT XII Pg.# 200

131) NCERT XII Pg.# 212

132) NCERT XII Pg.# 222

133) NCERT XII Pg.# 199

134) NCERT XII Pg. # 207

135)

NCERT XII Pg.# 198

136) NCERT Pg. # 42

137) NCERT, Pg # 51

138) NCERT Pg. # 43

139)

NCERT, Pg # 40

140)

NCERT, Pg # 43

141)

NCERT, Pg # 49

142) NCERT (XIth) Pg. # 101

143)

Module
Final Answer – Option 1

144) NCERT Pg # 82, 83

145) NCERT Pg # 82

146) NCERT Pg#201

147) NCERT Pg#201

148) NCERT Page No. 202

149) NCERT, Pg. # 201

150) NCERT, Pg. # 187, 190

151)

Module

152) NCERT, Pg. # 207, 208

153) NCERT-XI, Pg. # 212

154) NCERT, Pg. # 208, 209, 211, 212

155) NCERT Pg.#232

156)

NCERT, Pg # 231

157)

NCERT, Pg # 232

158)

NCERT, Pg # 241

159) NCERT Pg.#246

160) NCERT 337

161) NCERT XI Pg.#248

162) NCERT XI Pg.#245

163) NCERT Pg.#219,228

164) NCERT XI Pg.#219

165) NCERT Pg.#24,25,26

166) NCERT XII Pg. No. # 33

167) NCERT Pg. # 31

168) **Explain Question :** to identify ejaculatory duct in given diagram.
Concept : This question is based on male reproductive system.
Solution : The ejaculatory duct is formed by the union of the vas deferens and the duct from the seminal vesicle, which is here represented as "B".

169) NCERT(XII) Page#34/37(H) Fig:2.9

170)

NCERT-XII Pg. # 137

171) NCERT-XII, Pg. # 134-137

172) NCERT-XII, Pg. # 135

173) NCERT XII Pg. No. # 133

174) NCERT XII Pg. No. # 126

175) NCERT Pg.# No. 167, 169

176)

NCERT-XII Pg. # 167, 169, 173, 174

177) NCERT-XII Pg # 174

178) NCERT-XII, Pg. # 183, 184

179) NCERT XII Page No. 179, 180(E)

180) NCERT-XII, Pg. # 158